

Helix Constitution — Universal CLAUDE.md

Helix Constitution — Universal CLAUDE.md

Field	Value
Revision	2
Created	2026-05-14
Last modified	2026-05-20
Status	active
Status summary	Mirrored Constitution.md §11.4.73 (main-spec versioning + revision discipline) and §11.4.74 (submodule-catalogue-first discovery) into this CLAUDE.md per the new mandates' cross-document propagation requirement.
Issues	none
Issues summary	—
Fixed	§11.4.73 + §11.4.74 mirrors
Fixed summary	landed in lockstep with the Constitution.md additions.
Continuation	—

Table of contents

- [How inheritance works](#)
- [MANDATORY DEVELOPMENT PRINCIPLES](#)
- [MANDATORY ANTI-BLUFF COVENANT — END-USER QUALITY GUARANTEE](#)
 - [§11.4.1 — FAIL-bluffs are equally forbidden](#)
 - [§11.4.2 — Recorded-evidence requirement](#)
 - [§11.4.3 — Per-environment-topology test dispatch](#)
 - [§11.4.4 — Test-interrupt-on-discovery + retest-from-clean-baseline](#)
 - [§11.4.5 — Captured-evidence quality analysis](#)
 - [§11.4.6 — No-guessing mandate](#)
 - [§11.4.7 — Demotion-evidence rule](#)

- §11.4.8 — Deep-web-research-before-implementation
- §11.4.9 — Batch-source-fixes-before-rebuild
- §11.4.10 — Credentials-handling mandate
- §11.4.10.A — Pre-store credential leak audit (User mandate, 2026-05-17)
- §11.4.11 — File-layout discipline
- §11.4.12 — Auto-generated docs sync
- §11.4.13 — Out-of-band sink-side captured-evidence
- §11.4.14 — Test playback cleanup
- §11.4.15 — Item-status tracking
- §11.4.16 — Item-type tracking
- §11.4.17 — Universal-vs-project classification of new rules (User mandate, 2026-05-14)
- §11.4.20 — Subagent-driven-by-default mandate (User mandate, 2026-05-14)
- §11.4.18 — Script documentation mandate (User mandate, 2026-05-14)
- §11.4.19 — Fixed-document column-alignment mandate (User mandate, 2026-05-14)
- §11.4.21 — Operator-blocked status + self-resolution exhaustion (User mandate, 2026-05-14)
- §11.4.22 — Document-sync commit discipline (User mandate, 2026-05-14)
- §11.4.24 — Build-resource stats tracking mandate (User mandate, 2026-05-14)
- §11.4.25 — Full-Automation-Coverage Mandate (User mandate, 2026-05-15)
- §11.4.26 — Constitution-Submodule Update Workflow Mandate (User mandate, 2026-05-15)
- §11.4.31 — Submodule-Dependency-Manifest Mandate (User mandate, 2026-05-15)
- §11.4.32 — Post-Constitution-Pull Validation Mandate (User mandate, 2026-05-15)
- §11.4.30 — .gitignore + No-Versioned-Build-Artifacts Mandate (User mandate, 2026-05-15)
- §11.4.29 — Lowercase-Snake_Case-Naming Mandate (User mandate, 2026-05-15)
- §11.4.28 — Submodules-As-Equal-Codebase + Decoupling + Dependency-Layout Mandate (User mandate, 2026-05-15)
- §11.4.27 — No-Fakes-Beyond-Unit-Tests + 100%-Test-Type-Coverage Mandate (User mandate, 2026-05-15)
- §11.4.33 — Type-aware closure-status vocabulary (User mandate, 2026-05-15)
- §11.4.34 — Reopened-source attribution mandate (User mandate, 2026-05-15)
- §11.4.35 — Canonical-root inheritance clarity (User mandate, 2026-05-15)
- §11.4.36 — Mandatory install_upstreams on clone/add Mandate (User mandate, 2026-05-15)
- §11.4.37 — Fetch-before-edit mandate (User mandate, 2026-05-15)
- §11.4.38 — Installable-Asset Evidence Mandate (User mandate, 2026-05-17)
- §11.4.40 — Full-suite retest before release tag mandate (User mandate, 2026-05-17)
- §11.4.41 — Pre-Force-Push Merge-First Mandate (User mandate, 2026-05-17)
- §11.4.42 — Iteration-discipline mandate (User mandate, 2026-05-18)
- §11.4.43 — TDD-Fix-Discipline mandate (User mandate, 2026-05-18)

- §11.4.44 — Document revision header mandate (User mandate, 2026-05-18)
- §11.4.45 — Integration-status-doc maintenance mandate (User mandate, 2026-05-18)
- §11.4.46 — Validate-recent-work-before-post-flash-tests mandate (User mandate, 2026-05-18)
- §11.4.47 — Firebase data review mandate (User mandate, 2026-05-18)
- MANDATORY HOST-SESSION SAFETY (Constitution §12)
 - Forbidden — directly OR indirectly
 - Required safeguards for heavy scripts
 - §12.6 Memory-Budget Ceiling — 60% MAXIMUM
 - §12.10 Continuation document maintenance
- MANDATORY ABSOLUTE DATA SAFETY — ZERO RISK (Constitution §9)
- MANDATORY COMMIT & PUSH CONSTRAINTS
- MANDATORY TESTING CONSTRAINTS
- Code conventions
- When in doubt

This is the **base CLAUDE.md** imported by every project that includes the Helix Constitution submodule. Project-level CLAUDE.md may extend or tighten any rule by adding an explicit Override: <section> block — but MUST NOT weaken them.

Last revision: 2026-05-14

How inheritance works

A consuming project's root CLAUDE.md MUST start with a clearly-marked inheritance pointer:

```
## INHERITED FROM constitution/CLAUDE.md
```

All rules in ``constitution/CLAUDE.md`` (and the ``constitution/Constitution.md`` it references) apply unconditionally. The project-specific rules below extend them.

Claude Code supports the `@path/to/file` import syntax natively, so a consuming project can also write `@constitution/CLAUDE.md` at the top of its own CLAUDE.md and Claude Code will resolve it recursively. For agents that do not support `@imports`, the pointer-block pattern above ensures the inheritance is at least readable.

MANDATORY DEVELOPMENT PRINCIPLES

NO BLUFF. Every change ships with positive-evidence validation on the real target environment. A test that passes without exercising the user-visible behaviour is a critical defect.

(Constitution §7.1 + §11.4 — these references point to `constitution/Constitution.md`.)

CRITICAL: All code changes MUST follow these principles WITHOUT EXCEPTION:

1. **Solutions MUST NOT be error-prone.** Every fix must be robust, not introduce new failure modes. If a fix solves problem A but creates problem B, it is NOT acceptable. Test the fix against ALL existing functionality before committing.
2. **No blocking operations inside synchronized / shared-lock regions.** Long computations, network calls, or sleeps inside synchronized / Lock-held regions cause deadlocks or timeouts in other threads.
3. **Always consider concurrent callers.** Any method called from multiple threads must be safe for rapid consecutive calls.
4. **Test the fix, not just the symptom.** Verify the fix works AND doesn't break anything else.
5. **Anti-bluff is mandatory** (Constitution §7.1) — every runtime test MUST source the project anti-bluff helper, call at least one explicit user-visible action before any PASS, capture state delta, and assert positive evidence.
6. **Real captured evidence for audio/video** (Constitution §7.1 + §11.4.5) — features producing audio output validated via captured audio; features producing video output validated via captured frames + OCR or pixel-diff.

MANDATORY ANTI-BLUFF COVENANT — END-USER QUALITY GUARANTEE

Forensic anchor — verbatim user mandate (2026-04-28):

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

This is the historical origin of the project's anti-bluff covenant. Every test, every Challenge, every gate, every mutation pair exists to make the failure mode (PASS on broken-for-end-user feature) mechanically impossible.

Operative rule: the bar for shipping is **not** "tests pass" but **"users can use the feature."** Every PASS MUST carry positive evidence captured during execution that the feature works for the end user. Metadata-only PASS, configuration-only PASS, "absence-of-error" PASS, and grep-based PASS without runtime evidence are all critical defects regardless of how green the summary line looks.

Tests AND Challenges (HelixQA, integration suites, smoke tests, acceptance suites) are bound equally — a Challenge that scores PASS on a non-functional feature is the same class of defect as a unit test that does.

Canonical authority: constitution/Constitution.md §11.4 and its sub-sections §11.4.1 through §11.4.16.

Non-compliance is a release blocker regardless of context.

§11.4.1 — FAIL-bluffs are equally forbidden

A test that crashes for a script-internal reason (undefined variable under set -u, regex error, malformed assertion, missing argument) and produces a FAIL exit code is just as misleading as a PASS-bluff. Both let real defects ship undetected. Every test **MUST** fail **ONLY** for genuine product defects — script-bug failures must be fixed at the source layer (helper library, shared lib, test source), not patched in individual call sites.

§11.4.2 — Recorded-evidence requirement

A test that emits PASS without captured visual or audio evidence of the user-visible feature actually working on the screen the user would see is a §11.4 PASS-bluff. Every PASS for a user-visible feature **MUST** be cross-checked against captured recording + action timeline.

§11.4.3 — Per-environment-topology test dispatch

Tests that depend on environment topology **MUST** detect topology at test entry and dispatch the topology-appropriate variant. SKIP-with-reason is the correct fallback when the required topology is absent; PASS-by-default is forbidden.

§11.4.4 — Test-interrupt-on-discovery + retest-from-clean-baseline

The moment any defect is re-discovered, re-produced, or newly identified during a test cycle, the cycle **MUST** stop. Then: systematic debugging → fix at root cause → four-layer test coverage (pre-build / post-build / runtime / meta-test paired mutation) → full rebuild → re-deploy on every target → full retest from beginning.

§11.4.5 — Captured-evidence quality analysis

Audio: presence (RMS amplitude), channel count, sample rate + bit depth, glitch census, coexistence-artifact census. Video: presence (non-zero frame count), routing target, frame health (drops / jitter / freeze), obstruction census (OCR for hostile overlays), resolution + codec. Every check is required for every PASS.

§11.4.6 — No-guessing mandate

Forensic anchor — verbatim user mandate (2026-05-08):

"'LIKELY' is guessing, we **MUST NOT** have guessing, since it can be or may not be! No bluffing and uncertainty is allowed at any cost! We **MUST** always know exactly precisely what is happening exactly, in any context, under any conditions, everywhere!"

Forbidden vocabulary in tests / gates / status reports / closure narratives / commit messages when describing causes: likely, probably, maybe, might, possibly, presumably, seems, appears to, guess, seemingly, apparently, perhaps, supposedly, conjectured, and synonyms.

Either prove the cause with captured forensic evidence and state it as fact, OR explicitly mark UNCONFIRMED: / UNKNOWN: / PENDING_FORENSICS: with a tracked-task ID for follow-up.

§11.4.7 — Demotion-evidence rule

Demotion from a FAIL classification to a lower-severity classification requires positive evidence captured under the SAME CONDITIONS (same target, same build, same cycle position, same load profile) that originally exposed the defect. "I cannot reproduce in isolation" is a hypothesis, not a finding.

§11.4.8 — Deep-web-research-before-implementation

Before designing a non-trivial fix, perform deep web research: official docs, vendor technical guides, open-source codebases, coding tutorials, issue trackers. Every non-trivial fix's commit message (or accompanying entry) MUST cite at least one external source OR the literal "NO external solution found — original work".

§11.4.9 — Batch-source-fixes-before-rebuild

All source-side fixes that DO NOT require runtime validation to design MUST be landed BEFORE the next artifact rebuild. The anti-pattern of "fix A → rebuild → flash → cycle → fix B → rebuild → ..." serializes operator time onto rebuild latency.

§11.4.10 — Credentials-handling mandate

Forensic anchor — verbatim user mandate (2026-05-12):

| "Credentials or any secret and sensitive data MUST NOT leak!"

Credentials MUST NEVER be tracked in git. `.env` / `.env.*` / `*.env` patterns + `scripts/testing/secrets/*` (with `.example` + `README.md` exception) git-ignored project-wide. Test scripts MUST NEVER print or log credentials. Per-service file separation limits blast radius. `chmod 600` on credential files, `chmod 700` on parent directory. Rotation on suspected leak.

§11.4.10.A — Pre-store credential leak audit (User mandate, 2026-05-17)

Forensic anchor — verbatim user mandate (2026-05-17):

| "Us these for all future testing (full automation testing) and make sure they are not leaking anywhere or get git versioned!"

When an operator provides credentials, API tokens, signing keys, or any other secret material to be stored in the project's gitignored configuration, the storing agent MUST FIRST execute a repo-wide audit for prior leaks of THOSE specific values BEFORE storing: (1) `git ls-files | xargs grep -l <value>` for tree-leaks, (2) `git log -S<value> --all --source --remotes` for history-leaks, (3) surface findings to operator BEFORE storing (operator may rotate, accept-as-

compromise, or abort), (4) on finding open a §6/§7 sixth-law-incidents record + redact tracked files in-place to <redacted-per-§11.4.10> + record OPERATOR ACTION REQUIRED for rotation per §11.4.10 sub-clause 7, (5) extend pre-push hook credential-pattern grep to catch the escaped class in the same commit. See Constitution §11.4.10.A for the full mandate.

§11.4.11 — File-layout discipline

Project files organised by purpose, not historical accident. Source under canonical project roots. Tests under canonical test directories. Logs and forensic artifacts under operator-controlled directories — never scattered at repo root, never tracked unless they are reference assets.

§11.4.12 — Auto-generated docs sync

Every auto-generated document MUST be regenerated in the same commit as any edit to its source. All output formats (.md + .html + .pdf) MUST stay in sync at all times.

§11.4.13 — Out-of-band sink-side captured-evidence

Whenever a downstream consumer (HDMI sink, cloud monitor, downstream service) provides a network-accessible introspection API that reports what was actually received, the test suite MUST consume that report as captured-evidence. The on-source-side view alone is insufficient.

§11.4.14 — Test playback cleanup

Every test MUST leave the target in a quiescent state. Cleanup mandatory on every exit path (trap '<cleanup>' EXIT). The orchestrator MUST run a post-test sanity check and FAIL the just-completed test if it left orphan state.

§11.4.15 — Item-status tracking

Every active item in the project Issues file carries a ****Status:**** line within five lines of its heading. Six-state vocabulary: Queued, In progress, Ready for testing, In testing, Reopened, Fixed (→ Fixed.md). Status updated as the item progresses. All three Issues / Issues_Summary / Fixed file types kept in sync (Markdown + HTML + PDF).

§11.4.16 — Item-type tracking

Every active item in the project Issues file carries a ****Type:**** line within eight non-blank lines of its heading. Three-value CLOSED vocabulary: Bug (product defect / regression / user-visible broken behaviour), Feature (new capability not previously offered to end users), Task (internal workstream — refactor, doc, infra, gate, audit; the lowest-stakes default when ambiguous). The Issues_Summary file carries the Type column for every active item. All three Issues / Issues_Summary /

Fixed file types kept in sync (Markdown + HTML + PDF). Pre-build gates CM-ITEM-TYPE-TRACKING + CM-COVENANT-114-16-PROPAGATION enforce the mandate.

§11.4.17 — Universal-vs-project classification of new rules (User mandate, 2026-05-14)

Before adding ANY new rule, mandatory constraint, covenant clause, gate, or "MUST"-bearing statement to a project's Constitution / CLAUDE.md / AGENTS.md (or to a submodule's equivalents), the author MUST classify it as **universal** (reusable across any project → goes into this constitution submodule) or **project-specific** (references particular hardware / vendor / package / region → stays in the project / submodule layer). The commit message MUST carry a **Classification:** line stating the choice + one-sentence rationale. Universal rules that leak project-specific assumptions (hardware part numbers, vendor names, geographic regions, internal asset names) MUST be genericised first or downgraded to project-specific. When uncertain, default to project-specific (the narrower scope — lifting to universal later is cheap; the reverse is expensive). Pre-build gate CM-UNIVERSAL-VS-PROJECT-CLASSIFICATION audits new rule commits for the classification statement; paired mutation strips it and asserts gate FAILs.

§11.4.20 — Subagent-driven-by-default mandate (User mandate, 2026-05-14)

When the runtime supports subagent delegation (Claude Code Agent tool, Cursor task-runners, Aider sub-sessions, etc.), the primary agent MUST default to subagent delegation for any task that has multi-step scope (≥ 3 phases), parallelisable independent subtasks, long-running diagnostic loops, OR specialised domain workflows (code review, security audit, doc propagation). Foreground-only is reserved for single-file edits, mid-execution operator clarification, critical-state sequencing (commits / pushes / tags), or tasks so quick that subagent overhead exceeds the work. Sub-discipline: **tight scope** (4-6 tasks, not "do everything"), **checkpoint commits** after each major task, **anti-stall protection** explicit in prompts, **anti-bluff verification** of subagent claims via repo state. Parallel subagents MUST partition non-overlapping files; `commit_all.sh --auto-cascade` bundles both via `git add -A`. Gate CM-SUBAGENT-DELEGATION-AUDIT (when implemented in consuming project) flags foreground multi-step work as a §11.4.20 violation.

§11.4.18 — Script documentation mandate (User mandate, 2026-05-14)

Every Bash / shell / POSIX-sh script anywhere in a project (`scripts/`, `bin/`, `tests/`, library directories, deployment hooks, CI helpers — depth-N recursive) MUST carry: (1) an in-source documentation block (Purpose / Usage / Inputs / Outputs / Side-effects / Dependencies / Cross-references) at the top of the file; (2) an external user guide under `docs/scripts/<script-name>.md` covering Overview / Prerequisites / Usage examples / Edge cases / Internal behaviour / Related scripts / Last verified date. When a script is modified, BOTH the in-source block AND the external user guide MUST be updated in the SAME commit. **No documentation ever can be out of sync with its codebase.** Pre-build gate CM-SCRIPT-DOCS-SYNC walks every `*.sh` / `*.bash` under script directories, verifies a companion `docs/`

scripts/<name>.md exists, AND verifies doc was modified in the same commit (or doc-mtime ≥ script-mtime as a softer floor). Paired mutation strips the doc-sync invariant and asserts gate FAILs.

§11.4.19 — Fixed-document column-alignment mandate (User mandate, 2026-05-14)

Every project that maintains an open-work tracker AND a closed-archive tracker MUST keep the two structurally aligned along the same lifecycle axes (Status + Type). For the Fixed archive that means: (1) every ### / #### heading in Fixed.md (or equivalent) carries a ****Status:**** line and a ****Type:**** line within 8 non-blank lines of its heading; (2) a Fixed_Summary.md companion exists with the same column structure as Issues_Summary.md (# | Level | Status | Type | One-line description); (3) all three file formats (.md + .html + .pdf) for BOTH Fixed.md and Fixed_Summary.md stay in sync via the same single-shot wrapper that handles Issues + Issues_Summary; (4) closure migration is atomic — when an Issues entry resolves it moves to Fixed.md in the same commit, disappears from Issues_Summary (open-only), and appears in Fixed_Summary (closed-only). Status values for closed items are drawn from {Fixed (→ Fixed.md) | Fixed – pending device verification | Fixed – RECLASSIFIED}. Type values follow §11.4.16: {Bug | Feature | Task}. Pre-build gate CM-FIXED-COLUMN-ALIGNMENT (5+ invariants) — Fixed_Summary.md exists, table header carries Status+Type columns, mtime (Fixed_Summary ≥ Fixed), generator script present, sync wrapper invokes it, HTML+PDF exports for both Fixed and Fixed_Summary present. Paired mutation strips Status column from Fixed_Summary table header → gate FAILs. Classification: universal (per §11.4.17). No escape hatch.

§11.4.21 — Operator-blocked status + self-resolution exhaustion (User mandate, 2026-05-14)

Operator-blocked is the §11.4.15 Status closed-set's 7th value: {Queued | In progress | Ready for testing | In testing | Reopened | Operator-blocked | Fixed (→ Fixed.md)}. It is a **last-resort classification**, earned only after the agent documents exhaustion of every applicable self-resolution path: (a) CLI / ADB / SSH / API access already available, (b) subagent delegation per §11.4.20, (c) existing repo tooling (scripts / helpers / libraries), (d) captured fallback (synthetic event, asset substitution, mock, §11.4.3 topology SKIP), (e) external research per §11.4.8. Every Operator-blocked item MUST carry an ****Operator-Block-Details:**** line within 8 non-blank lines of its heading stating: **WHAT** (concrete action), **WHY** (each exhausted alternative enumerated), **UNBLOCK CONDITION** (observable signal), **WHO** (handle / contact / doc pointer). Issues_Summary.md lists Operator-blocked as a sortable Status value. Items MUST be re-evaluated every Nth tag cycle (project- defined, recommended ≥3rd cycle) — operator dependencies change. Fake Operator-blocked (no exhaustion audit) is a §11.4 covenant violation at the planning layer, severity-equivalent to a PASS-bluff. Gates: CM-ITEM-OPERATOR-BLOCKED-DETAILS (every Operator-blocked heading has the details line), CM-OPERATOR-BLOCKED-SELF-RESOLUTION- AUDIT (NEW Operator-blocked commits contain "Attempted: a — ...; b — ...; c — ..." trail). Classification: universal (per §11.4.17). No escape hatch.

§11.4.22 — Document-sync commit discipline (User mandate, 2026-05-14)

Every project tracking work items through an Issues / Fixed lifecycle MUST provide a **lightweight commit path** distinct from the full-repo commit wrapper. The lightweight path stages, commits, and pushes ONLY the status-tracking doc set — Issues + Issues_Summary + Fixed + Fixed_Summary + CONTINUATION + their HTML + PDF exports + any auto-generated audit artifact — so doc-status never drifts behind working-tree reality when the full-repo wrapper is unavailable (in-flight rebase, large submodule churn, partial network). The wrapper MUST: (a) auto-invoke the project's export-regeneration pipeline first so Markdown + HTML + PDF stay in sync; (b) stage ONLY the explicit doc-set list — NEVER `git add -A`; (c) use a separate flock disjoint from the full-tree wrapper's lock; (d) push to every parent-repo remote; (e) exit 3 on nothing-to-commit (informational, not error). The wrapper MUST be invocable standalone OR as a delegation flag on the full-tree wrapper (e.g. `commit_all.sh --docs-only`) so operators have a single mental model. Inherits the project's §9 preflight discipline (refuses to run mid-meta-test). Gate CM-COMMIT-DOCS-EXISTS verifies wrapper + guide + flag + doc-set enumeration. Paired mutation strips the doc-set array → gate FAILs. Classification: universal (per §11.4.17). Composes with §11.4.12 (export-sync), §11.4.15 (status tracking), §11.4.18 (script documentation), §12.10 (CONTINUATION maintenance). No escape hatch — doc-status drift is a §11.4 PASS-bluff at the documentation layer.

§11.4.24 — Build-resource stats tracking mandate (User mandate, 2026-05-14)

Every project under this Constitution with a build exceeding 1 minute wall-clock MUST run a host-side resource sampler for every build that captures memory used, CPU%, load average, disk read/write at a fixed interval (recommended 5 s) and computes per-metric **min / max / mean / p95** at stop. Per-build summaries appended to a TSV registry; the registry is the single source of truth — the human-readable Markdown report (and its HTML + PDF exports per §11.4.12) is derived. Top of the report MUST surface **ever-values** (min / max / mean across all tracked builds). Per-build entries sorted most-recent-first; each row carries SUCCESS / FAIL / UNKNOWN + reason for FAIL. Sampler MUST itself stay under 50 MB RSS and 5% CPU (Heisenberg-class observer constraint). Stop hook MUST be called from both success AND failure paths of the build wrapper. The Stats.{md,html,pdf} triple is committed via the project's §11.4.22 lightweight doc-sync wrapper. Gate CM-BUILD-RESOURCE-STATS-TRACKER + paired mutation hiding the monitor aside → gate FAILs. Classification: mixed (per §11.4.17) — the discipline universal, the implementation paths project-specific. Composes with §11.4.12 / §11.4.18 / §11.4.22 / §12.6 / §12.7 / §12.9 (the host-safety forensic anchors are the empirical motivation for this telemetry). No escape hatch — build-resource debugging without time-series data is the bluff this anchor forbids.

§11.4.25 — Full-Automation-Coverage Mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"Make sure that every feature, every functionality, every flow, every use case, every edge case, every service or application, on every platform we support is covered with full automation tests which will confirm anti-bluff policy and provide the proof of fully working capabilities, working implementation as expected, no issues, no bugs, fully documented, tests covered! Nothing less than this does not give us a chance to deliver stable product!"

For every consuming project, no feature / functionality / flow / use case / edge case / service / application on any supported platform may be considered **deliverable** until it is covered by automation tests proving six invariants: (1) anti-bluff posture (captured runtime evidence per §7.1 + §11.4); (2) proof of working capability end-to-end on the target topology (per §11.4.3, not in a mock); (3) working implementation matching the documented promise; (4) no open issues / bugs surfaced by the suite (cross-checked against §11.4.15 / §11.4.16 trackers); (5) full documentation (user manual entry + §11.4.18 for scripts) kept in sync per §11.4.12; (6) four-layer test floor per §1 (pre-build + post-build

- runtime + paired mutation). Consuming projects **MUST** publish a coverage ledger (feature × platform × invariant-1..6 × status), regenerated as part of release-gate sweeps, with gaps tracked per §11.4.15. Classification: universal (§11.4.17). No escape hatch. A project that ships a feature without all six invariants is **not delivering a stable product** — severity-equivalent to a §11.4 PASS-bluff at the release-gate layer. See Constitution §11.4.25 for the full mandate (cross-cutting reach, composition, audit requirements).

§11.4.26 — Constitution-Submodule Update Workflow Mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"Every time we add something into our root (constitution Submodule) Constitution, CLAUDE.MD and AGENTS.MD we **MUST FIRST** fetch and pull all new changes / work from constitution Submodule first! All changes we apply **MUST BE** committed and pushed to all constitution Submodule upstreams! In case of conflict, IT **MUST BE** carefully resolved! Nothing can be broken, made faulty, corrupted or unusable! After merging full validation and verification **MUST BE** done!"

Before ANY modification to constitution/Constitution.md, constitution/CLAUDE.md, or constitution/AGENTS.md, the agent or operator **MUST** execute the following pipeline in order:

1. **Fetch + pull first** — inside the constitution submodule worktree run `git fetch` against every remote, then `git pull --ff-only` (or `--rebase` if non-FF-mergeable; never `--strategy=ours` / `--allow-unrelated-histories` without explicit authorization). The submodule **MUST** be at upstream tip **BEFORE** any local edit.
2. **Apply the change** — classify per §11.4.17 (only universal additions belong here; project-specific clauses stay in the consuming project's governance). Cite the verbatim user mandate if one originated the change.
3. **Validate before commit** — run `meta_test_inheritance.sh` (or equivalent); verify no merge-conflict markers (`<<<<<<<, =====, >>>>>>>`); verify Constitution + CLAUDE + AGENTS cross-reference the new clause consistently.

4. **Commit + push to ALL upstreams** — stage only the governance files (NEVER `git add -A` inside the submodule); commit message cites the user mandate + §11.4.17 classification; push to every configured remote. A commit landing on one upstream but not others is a §2.1 violation AND a §11.4.26 violation.
5. **Conflict resolution** — if `pull --ff-only` reports non-fast-forward, merge carefully (preserve union of governance content, no clause silently dropped, re-classify, re-validate). Force-push to "make conflicts go away" is FORBIDDEN (§9.2). Nothing about the constitution may be broken, made faulty, corrupted, or rendered unusable.
6. **Post-merge validation + verification** — after the push lands, `git submodule update --remote --init` and re-run the consumer project's cascade verifier (per CONST-047) to confirm the new clause reaches every owned submodule. Any cascade gap closed in the same change-window.
7. **Bump consuming project pointer** — `.gitmodules`-tracked submodule pointer MUST be advanced to the new constitution HEAD in the SAME commit as any cascade work. Out-of-sync pointers are §11.4.26 violations.

Classification: universal (§11.4.17). No escape hatch. A constitution-submodule change violating §11.4.26 is a release blocker for every consuming project, severity-equivalent to a force-push without §9.2 authorization. See Constitution §11.4.26 for the full mandate (operational scope, cross-cutting reach).

§11.4.31 — Submodule-Dependency-Manifest Mandate (User mandate, 2026-05-15)

Every owned-by-us submodule MUST ship a machine-readable dependency manifest at canonical path `helix-deps.yaml` (or `.json/.toml`) listing its own-org Git SSH dependencies. Schema (per Constitution §11.4.31): `schema_version`, `deps: [{name, ssh_url, ref, why, layout: flat|grouped}], transitive_handling.recursive: true, transitive_handling.conflict_resolution: operator-required, language_specific_subtree: bool`.

Tooling: `incorporate-submodule <ssh-url>` adds the submodule at its declared canonical path (CONST-051(C) flat/grouped), reads its `helix-deps.yaml`, recurses for each declared dep, aborts on conflicting refs, emits `<root>/helix-manifest.yaml` audit record.

Anti-bluff guarantee: every manifest paired with a Challenge that bootstraps a throwaway consuming project, runs `incorporate-submodule`, asserts produced layout matches manifest, runs the submodule's own tests against the bootstrapped layout, captures wire evidence per §11.4.2. A manifest without this proof is a §11.4.31 violation.

§11.4.31 is the operational complement of §11.4.28 / CONST-051(C): nested own-org submodule chains are FORBIDDEN, manifests are the bridge that lets consumers reconstruct the dependency graph at the parent root.

Classification: universal (§11.4.17). Composes with §1, §3, §11.4.12, §11.4.17, §11.4.18, §11.4.20, §11.4.25, §11.4.26, §11.4.27, §11.4.28, §11.4.29, §11.4.30, CONST-047. See Constitution §11.4.31 for the full mandate.

§11.4.32 — Post-Constitution-Pull Validation Mandate (User mandate, 2026-05-15)

Whenever a project's constitution submodule is fetched + pulled with any content change, the project **MUST** run a full-project + recursive-submodule validation sweep **BEFORE** the new constitution HEAD is treated as canonical for any other work.

Sweep contract (canonical script: `scripts/verify-all-constitution-rules.sh`): re-runs the governance- cascade verifier; for every rule with a programmatic gate (CONST-053 `.gitignore` audit, CONST-051(C) nested-own-org-chain audit, CONST-052 case audit, CONST-050(A) mock-from-production audit, CONST-035 anti-bluff smoke), runs the gate against post-pull tree. Failures produce directed FAIL entries → tracker per §11.4.15 with Status: Reopened, Type: Bug. Closure requires positive-evidence per §11.4 anti-bluff covenant.

Pull-time invocation: `git submodule update --remote constitution` triggers the sweep automatically (post-update hook or commit wrapper); operator-explicit manual invocation also available.

Anti-bluff: sweep's own meta-test (paired mutation §1.1) plants a known violation of each enforced gate and asserts sweep reports FAIL for the planted gate. A sweep that exits PASS without running every implementable gate is a §11.4.32 violation.

§11.4.32 is the **enforcement engine** for every other §11.4.x and CONST-NNN rule — without it, new rules cascade as anchors but never get enforced in the codebase.

Classification: universal (§11.4.17). Composes with every rule that has a programmatic gate. See Constitution §11.4.32 for the full mandate.

§11.4.30 — .gitignore + No-Versioned-Build-Artifacts Mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"every project module, every Submodule, every service and application **MUST HAVE** proper `.gitignore` file! We **MUST NOT** git version build artifacts, cache files, tmp files, main `.env` file(s) or any files containing sensitive data, API keys or token! Any build derivative which we can recreate by executing proper mechanism for generating **MUST NOT** be versioned! We **MUST** pay attention what is going to be committed every time we are preparing to execute commit! If any violation is detected it **MUST** be fixed before commit is executed!"

Every project module / owned-by-us submodule / service / application **MUST** ship a proper `.gitignore` covering the forbidden-from-version-control classes:

1. **Build artefacts:** `/bin/`, `/build/`, `/dist/`, `/out/`, `target/`, `*.exe`, `*.dll`, `*.so`, `*.dylib`, `*.a`, `*.o`, `*.class`, `*.pyc`, generator-produced files (when the generator is committed).
2. **Cache files:** `__pycache__/`, `.pytest_cache/`, `.mypy_cache/`, `.ruff_cache/`, `node_modules/`, `.next/`, `.nuxt/`, `.cache/`, `.gradle/`, `.terraform/`, language-server caches.

3. **Temp files:** *.tmp, *.swp, *~, .DS_Store, Thumbs.db, *.orig, *.rej.
4. **Sensitive-data files:** .env, .env.* (allow .env.example placeholder), *.pem, *.key, *.crt, id_rsa*, id_ed25519*, .netrc, secrets/, api_keys.sh.
5. **Generated reports/logs:** *.log, coverage.out, htmlcov/, runtime captures unless reference assets.
6. **OS/IDE personal state:** .idea/, .vscode/ (except shared settings), .history/.

Anti-bluff invariant: .gitignore line alone is not sufficient — no file matching the forbidden patterns may be currently tracked. A tracked *.log despite the ignore-line is a violation of equal severity to no ignore-line at all.

Pre-commit attention: every commit author (human OR agent) **MUST** inspect `git diff --staged + git status` BEFORE the commit. Forbidden-class hits abort the commit until fixed (un-stage, add to .gitignore, scrub if already-tracked). Gate CM-GITIGNORE-PRECOMMIT-AUDIT + paired mutation.

Secret-leak intersection: §11.4.30 composes tightly with §11.4.10

- §12.1 (CONST-042) — a .env leak is BOTH a §11.4.30 and a §11.4.10 violation, requiring rotation + post-mortem.

Recreatable-content test: if a documented mechanism regenerates the file from sources, it's a build derivative and **MUST** be ignored. Generators **MUST** be committed so consumers regenerate on demand.

Classification: universal (§11.4.17). No escape hatch beyond enumerated exceptions. Severity-equivalent to §11.4 PASS-bluff at the repository-hygiene layer. Composes with §1, §2, §9.1, §11.4.10, §11.4.12, §11.4.17, §11.4.18, §11.4.20, §11.4.25, §11.4.26, §11.4.27, §11.4.28, §11.4.29, CONST-047. See Constitution §11.4.30 for the full mandate.

§11.4.29 — Lowercase-Snake_Case-Naming Mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"naming convention for Submodules and directories (applied deep into hierarchy recursively) - all directories and Submodules **MUST HAVE** lowercase names with space separator between the words of ' _ ' character (snake-case)! All existing Submodules and directories which are not following this rule **MUST BE** renamed! ... NOTE: Rules lowercase / snake-case do apply to all project files as well and references to it and from them!"

Every directory, submodule, and file **MUST** use lowercase snake_case names (ASCII letters / digits / underscores, words separated by _). Existing non-compliant names **MUST** be renamed as part of the migration window opened by this clause. Every reference (configs, docs, links, source-code imports, governance files) **MUST** be updated atomically with the rename — reference drift after a rename is a §11.4.29 violation of equal severity to the rename itself.

Exceptions (common-sense, must-not-break-technology). Language- mandated case (Java/Kotlin package paths, Android resource directories, Apple framework dirs, C#/Swift project layouts) is preserved inside the language-root. Submodule

root directory follows our convention; language-specific subtree follows its own. Vendor/upstream third-party submodules keep their upstream names. Build artefacts (node_modules/, __pycache__/, .git/, target/, build/, bin/) keep tool-mandated names. "Does renaming break the technology?" trumps the rule.

Upstreams/ → upstreams/ transition. Constitution submodule's `install_upstreams.sh` (exported via `.bashrc/ .zshrc`) MUST support BOTH Upstreams/ and upstreams/ directory layouts during migration. Lowercase wins when both present. Uppercase fallback retires only by deliberate amendment.

Project-Toolkit Upstreamable synchronisation. Upstreamable / Project-Toolkit machinery MUST be fetched+pullled before any rename batch + MUST itself comply with this rule. Lacking BOTH- directory support is a release blocker.

Test coverage of renames. Each rename batch ships with: (i) regression test verifying every reference now resolves; (ii) full CONST-050(B) test-type matrix run on the post-rename tree; (iii) anti-bluff wire-evidence captured. All three or it's a §11.4.29 violation.

Phased execution. Comprehensive brainstorming → phase-divided plan → fine-grained tasks/subtasks → every change covered by every applicable test type. Phases run in parallel with mainstream work (§11.4.20 subagent delegation).

Classification: universal (§11.4.17). No escape hatch beyond the common-sense exceptions enumerated. Severity-equivalent to §11.4 PASS-bluff at the reference-integrity layer. Composes with §1, §1.1, §11.4.12, §11.4.17, §11.4.18, §11.4.20, §11.4.25, §11.4.26, §11.4.27, §11.4.28, CONST-047. See Constitution §11.4.29 for the full mandate.

§11.4.28 — Submodules-As-Equal-Codebase + Decoupling + Dependency-Layout Mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"All existing Submodules in the project that we are controlling and belong to some our organizations (vasic-digital, HelixDevelopment, red-elf, ATMOSphere1234321, Bear-Suite, BoatOS123456, Helix-Flow, Helix-Track, Server-Factory — we can ALWAYS check dynamically using GitHub and GitLab CLIs) are equal parts of the project's codebase! We MUST work on that code as much as we do with main project's codebase! All on equal basis! Equally important! ... We MUST NEVER modify Submodules to bring into them any project specific context since they all MUST BE ALWAYS fully decoupled, project not-aware, fully reusable and modular (by any other project(s)), completely testable! All Submodule dependencies that are used by Submodule MUST BE accessed from the root of the project! We MUST NOT have nested Submodule dependencies but accessing each from proper location from the root of the project — directly from project's root `project_name/ submodule_name` or some more proper structure `project_name/submodules/ submodule_name`!"

Three cooperating invariants:

(A) Equal-codebase. Every owned-by-us submodule (orgs: vasic-digital, HelixDevelopment, red-elf, ATMOSphere1234321, Bear-Suite, BoatOS123456, Helix-Flow, Helix-Track, Server-Factory — dynamically discoverable via gh / glab CLIs) is an **equal part** of the consuming project's codebase. Same engineering attention as main: analysis, extension, test creation, gap-filling, bug-fix, documentation (user manuals, guides, diagrams, SQL, websites, all materials). A round that improves main while leaving an owned-submodule deficiency unaddressed is a §11.4.28 violation, severity-equivalent to a §11.4 PASS-bluff at the project-scope layer. Coverage ledgers (§11.4.25) list every owned submodule as in-scope.

(B) Decoupling / reusability. Owned submodules **MUST** stay fully decoupled, project-not-aware, reusable, modular, completely testable. **NEVER** inject project-specific context (hardcoded paths, hostnames, asset names) **INTO** a submodule. When a submodule needs parent-project info, use configuration injection (env var, config file, constructor parameter) — never a hardcoded reach.

(C) Dependency-layout. Every dependency consumed by an owned submodule **MUST** be accessible from the parent project's root at:

```
<project_root>/<submodule_name>/  
<project_root>/submodules/<submodule_name>/
```

Nested own-org submodule chains are FORBIDDEN. A submodule **MUST NOT** have its own .gitmodules entries pulling in further owned- by-us repos. Add the dependency at the parent's root path; the submodule reaches it via documented import / SDK / runtime resolver. Third-party submodules exempt.

Gates: CM-OWNED-SUBMODULE-EQUAL-ENGINEERING (release-gate sweep audits parity), CM-OWNED-SUBMODULE-DECOUPLING (pre-commit greps for parent-project context inside the submodule diff), CM-OWNED-SUBMODULE-LAYOUT (pre-merge verifies canonical location + no nested own-org submodules + dependency lookup from root). Paired mutations (§1.1) for all three. Classification: universal (§11.4.17). No escape hatch. Composes with §1, §3, §11.4.17, §11.4.20, §11.4.25, §11.4.26, §11.4.27, CONST-047. See Constitution §11.4.28 for the full mandate.

§11.4.27 — No-Fakes-Beyond-Unit-Tests + 100%-Test-Type-Coverage Mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"Mocks, stubs, placeholders, TODOs or FIXMEs are allowed to exist **ONLY** in Unit tests! All other test types **MUST** interact with real fully implemented System! No fakes, empty implementations or bluffing is allowed of any kind! All codebase of the project **MUST BE** 100% covered with every supported test type: unit tests, integration tests, e2e tests, full automation tests, security tests, ddos tests, scaling tests, chaos tests, stress tests, performance tests, benchmarking tests, ui tests, ux tests, Challenges (fully incorporating our Challenges Submodule). **EVERYTHING MUST BE** tested using HelixQA (fully incorporating HelixQA Submodule). HelixQA **MUST BE** used with all possible written tests suites (test banks) for every applications, service, platform, etc and execution of the full HelixQA QA autonomous sessions! All required dependency Submodules **MUST BE** added into the project as well (fully recursive!!!)."

Two cooperating invariants:

(A) No-fakes-beyond-unit-tests. Mocks, stubs, fakes, placeholders, TODO, FIXME, "for now", "in production this would", or empty-implementation patterns are PERMITTED only in unit-test sources. Every other test type — integration, E2E, full automation, security, DDoS, scaling, chaos, stress, performance, benchmarking, UI, UX, Challenges, HelixQA — MUST exercise the real, fully implemented system against real infrastructure. Production code MUST NOT import mock paths. Gate CM-NO-FAKES-BEYOND-UNIT-TESTS + paired mutation.

(B) 100% test-type coverage. Codebase MUST be covered by every supported test type the domain warrants: unit, integration, E2E, full-automation, security, DDoS, scaling, chaos, stress, performance, benchmarking, UI, UX, Challenges (vasic-digital/ Challenges submodule fully incorporated), HelixQA (HelixDevelopment/ HelixQA submodule fully incorporated). HelixQA autonomous sessions drive end-to-end execution of every registered test bank with captured wire evidence per check.

Required dependency submodules (recursive per CONST-047):

- Challenges — `git@github.com:vasic-digital/Challenges.git`
- HelixQA — `git@github.com:HelixDevelopment/HelixQA.git`
- Any other functionality submodules under `vasic-digital/* / HelixDevelopment/*` orgs the project depends on.

Pointers bumped to upstream HEAD in same commit as cascade work (§11.4.26 step 7); pointer drift = §11.4.27 violation.

Classification: universal (§11.4.17). Severity-equivalent to a §11.4 PASS-bluff at the release-gate layer. No escape hatch. Composes with §1, §7.1, §11.4.1–§11.4.26 (esp. §11.4.25 — this is its strict expansion into per-type-of-test territory). See Constitution §11.4.27 for full mandate.

§11.4.33 — Type-aware closure-status vocabulary (User mandate, 2026-05-15)

Every project that tracks work items by Type per §11.4.16 MUST close them with the Type-appropriate closure-status word, drawn from this 3-element closed map:

Item **Type:**	Closure **Status:** value
Bug	Fixed (→ Fixed.md)
Feature	Implemented (→ Fixed.md)
Task	Completed (→ Fixed.md)

The (→ Fixed.md) suffix is preserved across all three so the existing migration-discipline tooling (atomic Issues.md → Fixed.md move per §11.4.19) keeps working without per-Type branching. Generators (`generate_issues_summary.sh`, `generate_fixed_summary.sh`, status-counter helpers, the §11.4.23 colorizer) MUST treat the three terminal values as semantically equivalent (all map to "closed, positive evidence captured") while preserving the literal in the emitted document.

Closing a Feature with Fixed (→ Fixed.md) or a Task with Implemented (→ Fixed.md) is a §11.4.33 violation. Pre-build gate (recommended) CM-CLOSURE-VOCAB-TYPE-AWARE walks every Fixed.md heading + every Issues.md heading whose ****Status:**** is one of the three terminal values and asserts the Status-Type match. Composes with §11.4.15 (status tracking), §11.4.16 (type tracking), §11.4.19 (Fixed-document column alignment), §11.4.23 (colorisation). Classification: universal (per §11.4.17). No escape hatch.

§11.4.34 — Reopened-source attribution mandate (User mandate, 2026-05-15)

Every Issues.md (or equivalent project tracker) heading whose ****Status:**** is Reopened MUST carry, within 8 non-blank lines of the heading, a ****Reopened-Details:**** line capturing four sub-facts:

- **By:** AI or User (source-of-truth observer who flipped the status). AI covers in-loop reopens (test failure, gate regression, captured-evidence retrospect). User covers operator-side observations (manual testing, end-user report, design reconsideration).
- **On:** ISO date (YYYY-MM-DD).
- **Reason:** one-line cause classification — chosen from the closed vocabulary { test-failed | manual-testing-detected | captured-evidence-contradicts | end-user-report | cycle-re-discovered | design-reconsidered }. Other values are permitted with explicit Reason: <free text> annotation but the closed list MUST be tried first.
- **Evidence:** path to or short description of the captured artefact justifying the reopen — log file, recording, gate failure ID, operator quote, etc. Reopens without evidence are §11.4.6 / §11.4.7 violations: the reopen IS a demotion-from-Fixed classification change, and demotion requires positive evidence captured under the conditions that re-exposed the defect.

The Issues_Summary.md (or equivalent) Status column MUST distinguish the four Reopened sub-states by source so a sweep query for "reopens by AI in the last 30 days" is mechanically possible. Suggested column rendering: Reopened (AI: test-failed) vs Reopened (User: manual-testing).

A Reopened entry without ****Reopened-Details:**** is a §11.4.34 violation. Pre-build gate (recommended) CM-ITEM-REOPENED-DETAILS mirrors CM-ITEM-OPERATOR-BLOCKED-DETAILS (§11.4.21 walk pattern). Composes with §11.4.6 (no-guessing — Reason from closed vocabulary), §11.4.7 (demotion-evidence — reopen IS a demotion from Fixed), §11.4.15 (item-status tracking), §11.4.21 (Operator-blocked discipline — same audit-line pattern). Classification: universal (per §11.4.17). No escape hatch.

§11.4.35 — Canonical-root inheritance clarity (User mandate, 2026-05-15)

The constitution submodule's three files (constitution/Constitution.md, constitution/CLAUDE.md, constitution/AGENTS.md) ARE the canonical root — also called the parent files. They contain only universal rules per §11.4.17.

The consuming project's repository-root files (<project-root>/CLAUDE.md, <project-root>/AGENTS.md, optionally <project-root>/Constitution.md or equivalent) are consumer extensions. They open with the inheritance pointer (either the Claude-Code native @constitution/CLAUDE.md import or the portable ## INHERITED FROM constitution/CLAUDE.md heading defined in this file's "How inheritance works" section). They contain only project-specific rules per §11.4.17 — rules that reference particular hardware, vendor names, regulatory regions, internal asset names, or project-private conventions.

When in doubt about which file to edit: universal rule → edit constitution submodule's file; project-specific rule → edit consumer's file. Default consumer-side when uncertain (per §11.4.17, narrower scope is cheap to widen).

Terminology: when prose references "the parent CLAUDE.md" or "the root Constitution," the referent is the constitution-submodule file at constitution/<filename>, never the consumer's file. When it references "the project CLAUDE.md" or "this project's AGENTS.md," the referent is the consumer-side file at <project-root>/<filename>. AI agents resolve ambiguous references via this rule.

No silent demotion or silent promotion. Moving a rule between layers MUST be a visible commit — git mv of a section if it's a clean clone, or an explicit "Lifted from to constitution per §11.4.35" / "Demoted from constitution to per §11.4.35" line in the commit message.

Pre-build gate (recommended) CM-CANONICAL-ROOT-CLARITY verifies (a) consumer's CLAUDE.md opens with the inheritance pointer (either @import or ## INHERITED FROM constitution/CLAUDE.md heading), (b) the constitution submodule's three files are present at the expected path, (c) no ## INHERITED FROM block in the constitution submodule's own files (those ARE the source-of-truth, not consumers). Composes with §11.4.17 (universal-vs-project classification — §11.4.35 defines the file-layer split that §11.4.17 classifies INTO). Reading order: this anchor first, then §11.4.17. Classification: universal (per §11.4.17). No escape hatch.

§11.4.36 — Mandatory install_upstreams on clone/add Mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"Every Submodule or Git repository we add or clone MUST BE upstreams installed using Upstreamable utility which MUST BE available through exported paths of the host system (in .bashrc or .zhrc) using install_upstreams command executed from the root of the cloned (added) repository - only if in it is Upstreams or upstreams directory present with bash script files (recipes) for all repository's upstreams!"

Every clone / add of a Git repository under any consuming project MUST be followed by install_upstreams invocation from that repository's root IF its tree contains an upstreams/ directory (or legacy Upstreams/ per §11.4.29 transition) populated with *.sh recipe files declaring upstream Git SSH URLs.

`install_upstreams` is a host-system utility on operator's PATH (exported via `.bashrc/.zshrc`), implemented in this constitution submodule (`install_upstreams.sh`). The utility reads recipe files, configures every declared upstream as a named git remote, and fans out origin push URLs across all declared upstreams.

Skipping the invocation when `upstreams/` IS present silently breaks §2.1 (Multi-upstream push is the norm) — the next push lands on only one upstream. Gate CM-INSTALL-UPSTREAMS-ON-CLONE

- paired mutation (§1.1).

Automation: `incorporate-submodule` (§11.4.31) and `scripts/init-submodules.sh` patterns auto-invoke `install_upstreams` when applicable. Operator-explicit manual invocation remains supported.

Pre-commit attention: before the first commit in the newly-cloned working tree, verify `git remote -v | grep -c push` reports the expected upstream count.

Classification: universal (§11.4.17). Composes with §2, §2.1, §3, §9.2, §11.4.17, §11.4.20, §11.4.28, §11.4.29, §11.4.30, §11.4.31. See Constitution §11.4.36 for the full mandate.

§11.4.37 — Fetch-before-edit mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"Make sure that `feedback_fetch_before_edit` memory rule is part of our constitution Submodule - the root Consitution, AGENTS.MD and CLAUDE.MD. Validate and verify that Proejct-Toolkit and all Submodules do inherit all of them!"

The FIRST git-touching action of any session, on any consuming project, MUST be:

```
git fetch --all --prune
git log --oneline HEAD..@{u}          # parent
git submodule foreach --recursive 'git fetch --all --prune --quiet'
```

If `HEAD..@{u}` is non-empty, integrate (ff-merge / rebase / surface to operator per §11.4.4) BEFORE any local edit, scanner run, or test cycle. Multi-agent / multi-upstream codebases (Claude Code + Cursor + Aider + operator sessions in parallel) routinely lap each other; a 30-second fetch prevents the agent from redoing work a parallel session already finished, from filing a false-confidence "completion" of already-done work, and from doubling the multi-upstream conflict surface (§2.1) with sibling commits of the same change.

The check is non-negotiable even when the operator says "do X immediately" — skipping it on the basis of "nothing could have changed in the last N minutes" is a §11.4.6 (no-guessing) violation: remote state is not knowable without a fetch. The `fetch+log` output (even if empty) is the captured evidence.

Scope: consuming project root + every owned submodule recursively (§11.4.28) + the constitution submodule itself (§11.4.26 step 1 made this explicit for constitution-side edits; §11.4.37 generalises to ALL edits) + any dependency cloned via `incorporate-submodule` (§11.4.31) or `git submodule add` (§11.4.36).

Pre-build gate `CM-FETCH-BEFORE-EDIT-AUDIT` (when implemented in the consuming project) audits the most-recent commit range against the upstream HEAD at the commit's parent — non-aligned parent = FAIL. Paired mutation (§1.1): synthetic commit whose parent was N commits behind the then-current upstream HEAD → gate FAILs.

Classification: universal (§11.4.17). No escape hatch. Composes with §2.1, §11.4.4, §11.4.6, §11.4.20, §11.4.26, §11.4.32. See Constitution §11.4.37 for the full mandate.

§11.4.38 — Installable-Asset Evidence Mandate (User mandate, 2026-05-17)

For any user-distributable build artifact (package, bundle, installer, or container image produced by the build pipeline and distributed to end users), tests and challenges **MUST** open the artifact and verify each user-visible asset is **present** and **non-degenerate**.

A PASS without opening the artifact and verifying the asset chain end-to-end is a §11.4 PASS-bluff. The specific failure mode: source file exists → build packages it → post-build checks pass at the source layer → artifact produced with the asset stripped or misconfigured, and no gate ever opens the artifact to verify.

Each consuming project ships one challenge script per artifact type that opens the produced artifact and verifies every declared user-visible asset. The challenge **MUST** run as part of the project's standard QA gate.

Classification: universal (§11.4.17). No escape hatch. See Constitution §11.4.38 for the full mandate.

§11.4.40 — Full-suite retest before release tag mandate (User mandate, 2026-05-17)

A release tag **MUST NOT** be created until a **COMPLETE** retest with **ALL** existing tests has been executed on a clean baseline **AFTER** every workable item in the batch is done, fixed, polished, and individually verified. Spot-check retests that run only the tests touched by the batch are **FORBIDDEN** — they miss interaction defects between batch fixes and previously-stable code.

The complete retest comprises: (1) pre-build full sweep, (2) post-build full sweep, (3) on-device 4-phase cycle on **EVERY** owned device, (4) meta-test full mutation sweep, (5) Challenge bank full sweep, (6) Issues.md/Fixed.md state audit, (7) `CONTINUATION.md` sync check.

Time is essential — typically 12–48h elapsed effort. **NOT** optional, **NOT** abbreviated. Skipping is the exact "tests pass but feature broken" failure mode §11.4 prohibits.

Composes with §11.4.4 (per-fix retest still required at fix granularity) + §11.4.7 (full-suite retest is authoritative baseline for closures) + §11.4.39 (per-feature on-device validation runs as step 3 of the full-suite retest).

Classification: universal (§11.4.17). No escape hatch. See Constitution §11.4.40 for the full mandate.

§11.4.41 — Pre-Force-Push Merge-First Mandate (User mandate, 2026-05-17)

Forensic anchor — verbatim user mandate (2026-05-17):

"make sure we bring everything from branches to our side before forc push is done! Afer everything is safely and fully merged and all potential conflicts (if any) resolved, then do force push! make sure nothing isnlost, broken or corrupted on bith sides!"

Any force-push (`--force`, `--force-with-lease`, `+<ref>`, equivalent history-rewrite) authorised under CONST-043 MUST be preceded by a 4-step merge-first pipeline:

1. **Fetch every remote** — `git fetch --all --prune --tags` against origin + every upstream; capture output.
2. **Integrate every divergent commit locally** — `rebase / merge / operator-confirmed cherry-pick` per the appropriate strategy for every non-empty `HEAD..<remote>/<branch>` range.
3. **Audit the integrated tree** — no conflict markers anywhere (`grep -rn '^<====<<<<<< \|^===== $\|^>>>>>>'` returns empty in governance + source + test files); no file silently dropped; previously-passing tests still pass; captured- evidence artefacts still validate.
4. **Force-push** — only after steps 1-3 produce clean integration evidence: `git push --force-with-lease` (NEVER `--force` alone unless authorised per §9.2 sub-clause 6).

Two-gate composition with CONST-043. §11.4.41 does NOT relax CONST-043's operator-approval requirement — it adds a SECOND mechanical gate. Both required: CONST-043 alone authorises a push that loses remote work; §11.4.41 alone risks pushing without operator awareness.

Three failure modes prevented: (a) remote-side content loss when parallel sessions land work between fetches; (b) stale-state acts when `--force-with-lease` reads stale local refs; (c) conflict-driven corruption when markers get committed verbatim (observed 2026-05-17 in `helix_qa` + containers governance files).

Verification artefact — `docs/change logs/<tag>.md` "Force-push merge-first audit" section captures fetch output, per-remote divergence log, integration strategy, conflict- marker scan, test delta, push output with lease SHA, CONST-043 authorisation quote. Gate CM-FORCE-PUSH-MERGE-FIRST + paired mutation.

Classification: universal (§11.4.17). No escape hatch. Composes with §9.2, §11.4.4, §11.4.6, §11.4.26, §11.4.32, §11.4.37, §11.4.40, CONST-043, CONST-047. See Constitution §11.4.41 for the full mandate.

§11.4.42 — Iteration-discipline mandate (User mandate, 2026-05-18)

Project work proceeds in priority-ordered iteration cycles. Each cycle has five mandatory steps: (1) select TOP + MIDDLE critical items only (defer LOW until critical batch closed), (2) batch implementation with §11.4.4 four-layer coverage + §11.4.9 batch-source-fixes-before-rebuild, (3) smoke gate (<30 min — batch-touched tests + critical-path regression probe + anti-bluff baseline), (4) ONLY if smoke GREEN and no new operator/user report arrived, run §11.4.40 full retest (12–48 h), (5) release-ready OR loop back to step 1 with new evidence added to queue.

Composes with §11.4.4 (per-fix retest inside step 2), §11.4.7 (closures require same-conditions evidence; step 4 is authoritative baseline), §11.4.9 (source-side batching inside step 2), §11.4.34 (Reopened items attribute source), §11.4.40 (the multi-hour retest IS step 4 — §11.4.42 is the meta-loop conductor).

Anti-bluff coupling: every smoke and full-system PASS MUST carry positive captured evidence per §11.4.2 + §11.4.5. Tests AND HelixQA Challenges bound equally.

No escape hatch — no `--skip-priority-batch`, no `--skip-smoke`, no `--full-suite-only`. Subagents default to the §11.4.42 path.

Classification: universal (§11.4.17). See Constitution §11.4.42 for the full mandate.

§11.4.43 — TDD-Fix-Discipline mandate (User mandate, 2026-05-18)

Every fix MUST follow the 5-step TDD-fix workflow: **RED** (failing test FIRST, real product defect per §11.4.1, captured evidence per §11.4.2) → **LIVE-ADB-PROBE** (try fix on running device via `adb shell` for mutable surfaces — `setprop persist.*, settings put, pm clear, am ...`, boot-script push; INFEASIBLE for kernel / framework / HAL / vendor / init.rc / sepolicy / ro.* / Android.bp — must rebuild; cite `LIVE_PROBE_INFEASIBLE: <reason>` in commit message) → **GREEN** (source patch achieves same effect, batched per §11.4.9, four-layer coverage per §11.4.4) → **VERIFY** (re-run RED test, must PASS under SAME conditions per §11.4.7, captured positive evidence per §11.4.5, 10-iteration reliability per §11.4.42) → **DOCUMENT** (Issues.md → Fixed.md with type-aware closure vocabulary per §11.4.33, CLAUDE.md Applied Fixes row, changelog, guides, HelixQA bank, CONTINUATION.md per §12.10 — all in the SAME commit).

No escape hatch — no `--skip-red-test`, `--no-live-probe`, `--skip-verify` flag. "Test added after the fix" is a §11.4 PASS-bluff: it demonstrates the test agrees with the fix, not that the test catches the bug.

Classification: universal (§11.4.17). Composes with §11.4.1 / §11.4.2 / §11.4.4 / §11.4.5 / §11.4.7 / §11.4.9 / §11.4.40 / §11.4.42. See Constitution §11.4.43 for the full mandate.

§11.4.44 — Document revision header mandate (User mandate, 2026-05-18)

Every tracked document in scope (Issues.md, Issues_Summary.md, Fixed.md, Fixed_Summary.md, CONTINUATION.md, docs/guides/, **docs/research/**, docs/scripts/, **docs/changelogs/**, docs/superpowers/plans/, **docs/hardware/**, all other docs/* tracked Markdown) MUST carry a header block directly below the H1 title containing two MANDATORY fields: ****Revision:**** N (monotonic positive integer, never reset, never skipped) and ****Last modified:**** YYYY-MM-DDTHH:MM:SSZ (ISO 8601 UTC). Optional fields (Description, Authority, Maintainer, Scope) encouraged but not gated. CLAUDE.md / AGENTS.md / README / LICENSE / VERSION / rendered HTML / rendered PDF artifacts are explicitly OUT of scope (revision tracked via VERSION file or auto-derived from source Markdown).

Auto-bump: scripts/doc_revision_bump.sh <file> (idempotent), pre-commit hook for automatic staged-doc bumps, scripts/testing/sync_issues_docs.sh auto-bumps Issues_Summary / Fixed_Summary after regeneration, scripts/commit_docs.sh calls the bump before stage. CONTINUATION.md's existing Last updated: line per §12.10 IS the §11.4.44 Last modified: line — composed, not duplicated. HTML/PDF exports inherit revision from source Markdown via pandoc pipeline. No escape hatch — no --skip-revision-bump flag exists anywhere.

Pre-build gates CM-DOC-REVISION-HEADER-PRESENT + CM-COVENANT-114-44-PROPAGATION + paired mutations per §1.1. Composes with §12.10 (CONTINUATION.md header reuse), §11.4.12 (Issues_Summary regen), §11.4.22 (commit_docs.sh hook entry), §11.4.23 (HTML colorizer preserves revision), §11.4.18 (companion doc for doc_revision_bump.sh).

Classification: universal (§11.4.17). See Constitution §11.4.44 for the full mandate.

§11.4.45 — Integration-status-doc maintenance mandate (User mandate, 2026-05-18)

Every non-trivial domain integration MUST have a docs/<domain>/<integration>/Status.md document that (1) exists when more than one fix/test/gate has landed for the integration, (2) carries the §11.4.44 revision header, (3) is auto-synced (HTML

- PDF) on every related test cycle and every fix touching the integration, (4) is auto-colored per §11.4.23, (5) has a sync wrapper invocable as bash scripts/testing/sync_integration_status.sh or a per-integration thin shell, (6) lives under docs/<domain>/<integration>/, (7) includes a captured-evidence- driven status table per §11.4.5 (every claim cites the test log / recording / sink-probe report that backs it), (8) uses the closed status vocabulary PASS / FAIL / SKIP / PENDING_FORENSICS / OPERATOR-BLOCKED, (9) lists operator-blocked items at the top so operators find action items in O(1), (10) is referenced from docs/CONTINUATION.md §3 (Active work) when any item is non- terminal.

§11.4.45 is the generic form of §12.10 (CONTINUATION.md) applied to every integration domain. Without this generalisation each new integration re-invents the same sync wrapper, revision-header discipline, captured-evidence requirement, and operator-blocked surface.

Pre-build gates CM-COVENANT-114-45-PROPAGATION + CM-AF-INTEGRATION-STATUS-DOCS + paired mutations (propagation strip / Revision-line delete / sync-staleness / vocabulary violation). Composes with §11.4.5 (captured evidence), §11.4.12 (sync wrapper pattern reused), §11.4.13 (sink-side evidence is a specific instance), §11.4.15 (status vocabulary), §11.4.22 (commit_docs.sh wrapper), §11.4.23 (colorizer), §11.4.44 (revision header), §12.10 (CONTINUATION.md references Status.md paths).

No escape hatch — no --skip-status-sync, no --no-revision-bump-on-status, no --allow-stale-html flag.

Classification: universal (§11.4.17). See Constitution §11.4.45 for the full mandate.

§11.4.46 — Validate-recent-work-before-post-flash-tests mandate (User mandate, 2026-05-18)

After every device flash, the orchestrator MUST first run a recent- work validation pass (targeted on-device tests for items currently in Issues.md In progress / Ready for testing / Reopened, Fixed.md items closed within last 7 days, CONTINUATION.md §3 active work). Only if that pass is 100% green does the orchestrator proceed to the full post-flash suite.

Helper: scripts/testing/recent_work_validate.sh --device <serial> writes /data/local/tmp/.recent_work_validated IFF green; the full suite (test_all_fixes.sh) refuses to start without it. Marker is invalidated on reboot (stores device boot epoch).

Composes with §11.4.4 (STOP-on-discovery) + §11.4.6 (no-guessing) + §11.4.7 (demotion-evidence) + §11.4.40 (full-suite gate) + §11.4.42

- §11.4.43 + §11.4.44 + §12.10. Each recent-item fix MUST have a paired §11.4.43 RED-then-GREEN — a GREEN with no prior RED is a bluff.

Pre-build gates CM-COVENANT-114-46-PROPAGATION + CM-AF-RECENT-WORK-VALIDATION-GATE + CM-AF-VALIDATION-ARTIFACT-FILE + paired mutations.

Classification: universal (§11.4.17). No escape hatch — no --skip-validation, --full-suite-always, --ignore-recent-work flag. See Constitution §11.4.46 for the full mandate.

§11.4.47 — Firebase data review mandate (User mandate, 2026-05-18)

Before every "bigger working round" (pre-build, pre-flash, pre-tag, daily, post-deployment burn-in) the operator/loop MUST execute scripts/firebase/review_round.sh. The pass queries Crashlytics (fatals + non-fatals + ANRs) + Analytics + Performance, classifies findings by severity, dedup-maps to existing Issues.md entries via a three-tier algorithm (exact Firebase Issue-ID match →

stacktrace- similarity cluster hash → operator merge review), and drafts new Issue entries for unrecognised findings with full Firebase Console URL refs + §11.4.4(a) systematic-debugging output. Skipping the pass is a §11.4 PASS-bluff — Firebase IS the captured evidence from real end-user devices.

Five mandatory elements: (1) 5-trigger cadence (pre-build / pre- flash / pre-tag blocking, daily / burn-in non-blocking), (2) 3-source query (all three sources), (3) Issues.md output with Firebase metadata (Issue IDs + URL + Cluster Hash / KPI / Funnel), (4) 3-tier dedup, (5) comprehensive systematic-debugging output per Issue.

Pre-build gates CM-COVENANT-114-47-PROPAGATION + CM-AF-FIREBASE-REVIEW-CADENCE + CM-AF-FIREBASE-ISSUE-XREF + 3 paired mutations. Composes with §11.4.4 / §11.4.4(a) / §11.4.6 / §11.4.7 / §11.4.10 / §11.4.12 / §11.4.14 / §11.4.15 / §11.4.16 / §11.4.34 / §11.4.42 / §11.4.43 / §11.4.44 / §11.4.45 / §11.4.46.

No escape hatch — no --skip-firebase-review, --firebase-review-not-applicable, --no-issue-from-firebase flag. Operator MAY filter with --severity-min but MUST execute the pass.

Classification: universal (§11.4.17). See Constitution §11.4.47 for the full mandate.

§11.4.48 — UI-driven video testing mandate (User mandate, 2026-05-18)

Every test that asserts video playback on a secondary display MUST traverse the user-equivalent UI path (launcher icon → app home → content list → tile tap → playback → in-app pause/resume → back button stop). NOT Intent/Broadcast shortcuts (am start -a VIEW, cmd media_session play). Real uiautomator dump element resolution

- input tap X Y. Coverage: every video-capable app in PRODUCT_PACKAGES + every stream type (progressive HTTP / HLS / DASH / RTMP / file-local / DRM) + every codec (H.264/265/VP9/AV1/ MPEG-2/MP4V video; AC-3/E-AC-3/TrueHD/DTS/DTS-HD/MLP/Opus/AAC audio). Secondary display verified via ffprobe-on-captured-mp4 + VOM activeDecoder state. Arvus codec-state cross-check per §11.4.13 + §CG screenshot.

Per §11.4.4 four-layer: pre-build gate CM-AF-UI-DRIVEN-VIDEO- COVERAGE + propagation gate CM-COVENANT-114-48-PROPAGATION + on-device test framework at device/rockchip/rk3588/tests/ui_driven/ (Layer 1 helper + Layer 2 per-app drivers + Layer 3 scenarios) + Layer 4 orchestrator scripts/testing/ run_ui_driven_video_suite.sh + paired meta-test mutations.

Carve-out (User mandate 2026-05-20). The 5 canonical tracker documents — docs/Issues.md, docs/Issues_Summary.md, docs/Fixed.md, docs/ Fixed_Summary.md, docs/CONTINUATION.md — sit at docs/ root by design. They are architectural constants of the project layout, analogous to AOSP's Makefile, Android.bp, OWNERS files at repo root. Their location is encoded as literal path strings in §11.4.12 + §11.4.15 + §11.4.16 + §11.4.19 + §11.4.44 + §11.4.53 propagation gates plus the helper-script constellation that regenerates them. Moving them would require coordinated amendment of those 6 sister anchors plus 5 pre-build gates plus ~20 helper scripts plus 42 consumer files in a single PWU. Per §11.4.66, that scope is operator-blocked until explicitly authorised. Audit-snapshot files (docs/audit/anti_bluff_audit.md, docs/audit/

PRE_SONOS_TAG_READINESS.md, docs/audit/
D1_WIFI_FAIL_CLASSIFICATION.md, plus any future audit snapshots) DO move
under docs/audit/ per the §11.4.11 general principle.

Canonical authority: constitution submodule Constitution.md §11.4.48.

Non-compliance is a release blocker regardless of context.

§11.4.49 — Dual-approach testing mandate (User mandate, 2026-05-18)

Every feature test exercising a user-visible behaviour MUST ship in TWO variants:
a UI-driven variant (uiautomator-based, §11.4.48 surfaces A–E) AND an Intent/
Broadcast-driven variant (am start --es / am broadcast-based). Either alone is a
§11.4 PASS-bluff for the OPPOSITE half of the stack — UI catches app-side bugs;
Intent catches framework/system-server bugs.

Shared assertion base at tests/lib/dual_approach_test_base.sh: dat_init /
dat_start_capture / dat_assert_codec_state / dat_assert_video_frames /
dat_assert_audio_channels / dat_arvus_dashboard_capture /
dat_cleanup / dat_report_finding. Both variants gather identical evidence
into mirror directories qa-results/dual_approach/<F>/<run-ts>/
{ui,intent}/ so the orchestrator diffs results and pinpoints which half of the stack
contains a bug.

Kinopoisk 5.1 EAC3 is the canonical first implementation. Both variants are RED
per §11.4.43 until the §CN decoder pipeline fix lands.

Pre-build gates: CM-COVENANT-114-49-PROPAGATION (anchor across parent + 42
consumer files) + CM-AF-DUAL-APPROACH-COVERAGE (shared base contract) + CM-
AF-KINOPOISK-5-1-DUAL-COVERAGE (both variants exist + share the base). Three
paired meta-test mutations.

No escape hatch — no --ui-only / --intent-only / --skip- dual flag.

Canonical authority: constitution submodule Constitution.md §11.4.49.

Non-compliance is a release blocker regardless of context.

§11.4.50 — Deterministic consistency mandate (User mandate, 2026-05-18)

Every test that PASSes MUST have been executed N times (default N=3 normal
tests, N=10 cycle-validation suites) against the same firmware MD5 + same device
+ same topology and produced IDENTICAL PASS in every iteration. A divergent N-
iter run is auto-FAIL — there is no "first PASSED therefore X was a flake" path.
§11.4.7's intermittent / transient / flake / flaky vocabulary is enforced
MECHANICALLY by this mandate, not merely textually.

Coverage: every public API path (Activity / Service / Broadcast receiver /
ContentProvider URI / IPC interface / JNI entry / sysprop write / sysfs node / init.rc
trigger) MUST have ≥1 dedicated test that drives it. Untested paths surface in a
feature-coverage-matrix audit; the threshold ratchets 70 → 85 → 95 → 99 over
phases.

Reliability mechanism: `ab_run_n_times <test_name> <N> <fn> [args...]` helper in the project anti-bluff library loops, captures evidence-hash per iter, asserts all N hashes + exit codes identical. NO operator-facing escape converts divergence to PASS.

Pre-build gates: `CM-COVENANT-114-50-PROPAGATION + CM-AF-RELIABILITY-CHECK-WIRED + CM-AF-FEATURE-COVERAGE-MATRIX`. Three paired meta-test mutations. No escape hatch — no `--allow-flake`, `--first-pass-suffices`, `--skip-n-iter`, `--skip-coverage-audit` flag.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.50.

Non-compliance is a release blocker regardless of context.

§11.4.51 — Live-ADB-First Maximization Mandate (User mandate, 2026-05-18)

Every fix MUST be classified by rebuild-requirement before commit using the project's per-file-class decision matrix. If `LIVE_ADB_TESTABLE` (on-device test scripts, host scripts, atmosphere- *.sh boot scripts, persist.* properties, markdown docs, test fixture assets), the operator MUST first apply the fix to the running device via `adb push / setprop / pm install -r / mount -o remount, rw`, run the §11.4.43 RED test live, capture PASS, THEN commit + rebuild + reflash as belt-and-suspenders re-validation. Commit footer: `LIVE_ADB_VALIDATED: yes`. If `REQUIRES_REBUILD` (kernel, framework Java/AIDL, native C++ in APEX, sepolicy, `init.rc`, `ro.*` properties, XML overlays, codec XML in APEX, `Android.bp/.mk`), the operator proceeds directly to source-side + rebuild. Commit footer: `REQUIRES_REBUILD: <reason>`. Mixed batches use partial.

§11.4.51 REFINES §11.4.43 step 2 with mechanical enforcement. Helper: `scripts/testing/classify_fix_rebuild_requirement.sh` walks `git diff --name-only`, looks up each file against the matrix, emits per-file classification + recommended commit-message footer. Unmatched paths classify as `REQUIRES_REBUILD: unmatched-path` (safe default per §11.4.6). Pre-build gates: `CM-COVENANT-114-51-PROPAGATION + CM-AF-CLASSIFY-FIX-HELPER-EXISTS`

- `CM-AF-LIVE-ADB-FIRST-COMMIT-MARKER`. Three paired meta-test mutations. No escape hatch — no `--skip-classify` / `--assume-rebuild` / `--no-footer-required` flag.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.51.

Non-compliance is a release blocker regardless of context.

§11.4.52 — Autonomous-Validation Mandate (User mandate, 2026-05-18)

Forensic anchor — verbatim user mandate (2026-05-18):

"Make sure we have full automation tests which will do all this work in full automation! IMPORTANT: Make sure that all existing tests and Challenges do work in anti-bluff manner — they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

Every user-facing feature MUST have at least one autonomous validation path: end-to-end via `adb shell` + scripted automation, captured runtime evidence per §11.4.5, PASS/FAIL verdict WITHOUT a human present to drive UI, observe screen, or make decisions. Operator-attended tests are SUPPLEMENTARY, never PRIMARY. A feature whose ONLY validation path is operator-attended is a §11.4.52 violation — the path does not scale to CI, does not run on every commit, does not survive operator unavailability, and produces the exact "tests pass but feature doesn't work for users" failure mode §11.4 forbids.

Acceptable autonomous paths: programmatic instrumentation APK (SDK-API exercises like `MediaCodec.createDecoderByName` + JSON result file), headless intent dispatch + state poll (`am start --es`

- `dumpsys / /proc/<pid>/maps / media.metrics` polling), ADB-driven uiautomator (ONLY if `uiautomator dump | grep -c clickable=true ≥ 1` — near-empty hierarchy proves UI-driven INFEASIBLE and demands fallback to APK/intent), network-side sink probe (Arvus dashboard, Sonos REST, etc. per §11.4.13), HelixQA autonomous QA session (§11.4.27).

Per-feature coverage ledger (§11.4.25) MUST classify each row as AUTONOMOUS_VERIFIED / AUTONOMOUS_DESIGNED / OPERATOR_ATTENDED_ONLY / NOT_APPLICABLE. OPERATOR_ATTENDED_ONLY is a release blocker until promoted, citing a tracked migration work item per §11.4.15 + §11.4.16. Autonomous paths themselves MUST be anti-bluff: positive captured evidence per §11.4.5, paired meta-test mutation per §1.1.

Composes with §11.4.25 (full-automation-coverage — §11.4.52 strictens invariants 1+2), §11.4.27 (no-fakes-beyond-unit + 100% type coverage — operational layer closing the gap), §11.4.39 (per-feature on-device end-user validation — refined to mandate autonomous drivability), §11.4.43 (TDD-fix — autonomous path RED before fix, GREEN after), §11.4.48 (UI-driven — fallback to APK/intent when uiautomator hierarchy empty), §11.4.49 (dual-approach — Intent variant often IS the autonomous path), §11.4.50 (deterministic consistency — autonomous paths scale to N iterations), §11.4.51 (live-ADB-first — instrumentation APK is LIVE_ADB_TESTABLE).

Pre-build gates CM-COVENANT-114-52-PROPAGATION (anchor literal across canonical files) + CM-AF-AUTONOMOUS-PATH-PER-FEATURE (coverage-ledger classification column non-empty + valid value). Paired mutations strip the anchor literal AND inject an OPERATOR_ATTENDED_ONLY row without a tracked migration item. No escape hatch — no `--allow-operator-attended-only`, `--skip-autonomous-path`, `--manual-validation-suffices` flag.

Canonical authority: constitution submodule Constitution.md §11.4.52.

Non-compliance is a release blocker regardless of context.

§11.4.53 — Fixed_Summary parity mandate (User mandate, 2026-05-18)

Forensic anchor — verbatim user mandate (2026-05-18T17:55Z):

"Note: Just like for Issues we have Issues_Summary, for Fixed we MUST HAVE Fixed_Summary - like all other docs: ALWAYS in sync and up to date and ALWAYS exported into the PDF and HTML! Add this mandatory rule / constraint into the root (constitution Submodule) Constitution, AGENTS.MD and CLAUDE.MD."

docs/Fixed_Summary.md is the symmetric short-form summary of docs/Fixed.md. MUST be regenerated whenever Fixed.md changes. HTML + PDF exports MUST travel with the markdown (identical mtime within sync_issues_docs.sh granularity). Stale exports are §11.4.53 violations regardless of whether the underlying .md is correct — an operator (or future agent) reading the HTML or PDF gets a divergent view of which items are closed, and the §12.10 CONTINUATION resumption guarantee silently breaks. Same discipline as §11.4.12 Issues_Summary applied to Fixed.md.

Generator: scripts/testing/generate_fixed_summary.sh (canonical, MUST be executable, MUST emit a markdown table whose header columns include Status and Type per §11.4.19 column-alignment). Auto- sync wrapper: scripts/testing/sync_issues_docs.sh regenerates BOTH Issues_Summary AND Fixed_Summary in one shot (stages 1a + 1b), then exports HTML + PDF (stage 2), then colorizes per §11.4.23 (stage 3), then re-renders the PDFs from the colorized HTML (stage 3b). MUST be invoked after any edit to Fixed.md. No --issues-only flag exists, and §11.4.53 prohibits adding one.

Sort order: closure date DESC (most-recent-Fixed first), §-letter / Fix-# secondary. Documented at the top of the generated file.

Composes with §11.4.12 (Issues_Summary parity is the sibling rule; §11.4.12 + §11.4.53 are the canonical pair), §11.4.19 (atomic Issues→Fixed migration triggers Fixed_Summary regen — column-aligned structure), §11.4.23 (visual-cue + grouping colorizer post-processes both summaries), §11.4.33 (type-aware closure vocabulary — Fixed_Summary respects Fixed (→ Fixed.md) / Implemented (→ Fixed.md) / Completed (→ Fixed.md) terminal values literally), §11.4.44 (revision header applies to Fixed_Summary.md), §12.10 (CONTINUATION.md resumption guarantee depends on divergent-summary problem NOT existing).

Pre-build gates CM-FIXED-SUMMARY-SYNC (6 invariants: Fixed_Summary exists; HTML + PDF mtime ≥ md mtime; Fixed_Summary mtime ≥ Fixed mtime; generator script exists + executable; sync wrapper invokes generator) + CM-COVENANT-114-53-PROPAGATION (anchor literal across canonical files + per-consumer propagation). Paired mutations strip the anchor literal AND move the generator aside AND backdate Fixed_Summary mtime. No escape hatch — no --skip-fixed-summary-sync, --issues-only, --summary-not-applicable flag.

Canonical authority: constitution submodule Constitution.md §11.4.53.

Non-compliance is a release blocker regardless of context.

§11.4.54 — ATM-NNN ticket identifier mandate (User mandate, 2026-05-19)

Forensic anchor — verbatim user mandate (2026-05-19):

"Every workable item in Issues.md / Issues_Summary.md / Fixed.md / Fixed_Summary.md MUST carry a stable, unique, auto-incremental ATM-NNN ticket identifier. ATM- prefix, monotonic, never renumbered, append-only."

Every workable item in docs/Issues.md AND docs/Fixed.md MUST carry a [ATM-NNN] ticket identifier in its heading (form ## \$X.Y. [ATM-NNN] <title>, zero-padded ≥ 3 digits). Identifiers are allocated by scripts/testing/assign_atm_ticket_ids.sh and persisted to an append-only state file scripts/testing/.atm_ticket_state.json (jsonl: atm_id, heading_hash, type, current_location, current_status, reopens_count, created_at, last_modified). Once assigned, an ATM-NNN MUST NEVER be renumbered, reused, decremented, or skipped. heading_hash is the binding key — wording reflows preserve the binding via the state-file lookup.

Issues_Summary.md and Fixed_Summary.md MUST carry an ATM ID column as the leftmost data column. Generators (generate_issues_summary.sh, generate_fixed_summary.sh) emit the column so operators / agents can sort + filter on it.

Composes with §11.4.15 (Status), §11.4.16 (Type), §11.4.19 (column-alignment), §11.4.33 (closure vocabulary), §11.4.12 + §11.4.53 (Issues_Summary + Fixed_Summary regen — helper invoked from sync_issues_docs.sh), §11.4.55 (per-item Reopens.md path key), §11.4.57 (README.md doc-link cross-reference key).

Pre-build gates CM-ATM-TICKET-IDS-COMPLETE (every heading carries [ATM-NNN]) + CM-ATM-TICKET-IDS-UNIQUE (no duplicates) + CM-ATM-TICKET-IDS-MONOTONIC (no gaps) + CM-COVENANT-114-54-PROPAGATION (anchor literal across canonical files). Paired mutations strip an [ATM-NNN] heading token, dup an ID in the state file, gap the sequence at NNN=2, strip the anchor literal. No escape hatch — no --skip-atm-assignment, --renumber, --no-atm-id-required flag.

Canonical authority: constitution submodule Constitution.md §11.4.54.

Non-compliance is a release blocker regardless of context.

§11.4.55 — Reopens-history tracking + per-item Reopens.md doc (User mandate, 2026-05-19)

Forensic anchor — verbatim user mandate (2026-05-19):

"Add a Reopens-count column. For any item whose reopens-count > 0, create docs/issues/ATM-NNN/Reopens.md (+ HTML + PDF) with comprehensive reopen history + Fixed cycles. Each reopen MUST include By (AI / User), On, Reason, Evidence, and each Fixed-marking with reasoning chain."

Every workable item with reopens_count > 0 MUST have a companion document at docs/issues/<ATM-NNN>/Reopens.md (+ HTML + PDF). The document MUST contain: (1) §11.4.44 revision header, (2) item identification (ATM ID, Title, Type, Current Status, Current Location, link back to live heading), (3) cycle counters (Total reopens, Total fixed cycles), (4) chronological timeline — one entry per state-change event, each with By (AI/User per §11.4.34), On (ISO date), Event (Opened/Reopened/Fixed/Implemented/ Completed), Reason (closed-vocabulary value from §11.4.34 for reopens; captured-evidence summary for

closures), Evidence (path or short description), Outcome, (5) reasoning chain for each closure (root-cause analysis, captured-evidence under same conditions per §11.4.7, gate / mutation pair), (6) most-recent state-change pointer.

Issues_Summary.md and Fixed_Summary.md MUST carry a Reopens column; cells with count > 0 hyperlink to the per-item Reopens.md. The §11.4.23 colorizer MAY apply a visual cue when reopens > 2.

Composes with §11.4.34 (per-event capture in current heading — §11.4.55 is the per-item history aggregation), §11.4.54 (ATM-NNN provides the stable path), §11.4.44 (revision header), §11.4.45 (Status.md per-integration analogue), §11.4.53 (Fixed_Summary parity — Reopens column symmetric on both summaries).

Pre-build gates CM-REOPENS-DOC-EXISTS-WHEN-COUNT-GT-ZERO + CM-REOPENS-DOC-REVISION-HEADER + CM-REOPENS-COL-IN-SUMMARIES + CM-COVENANT-114-55-PROPAGATION. Paired mutations delete a Reopens.md for a reopens_count=2 item, strip the revision header, remove the column from Issues_Summary, strip the anchor literal. No escape hatch — no --skip-reopens-doc, --collapse-history, --reopens-not-applicable flag.

Canonical authority: constitution submodule Constitution.md §11.4.55.

Non-compliance is a release blocker regardless of context.

§11.4.56 — Status_Summary parity + two-audience format (User mandate, 2026-05-19)

Forensic anchor — verbatim user mandate (2026-05-19):

"Every Status.md doc gets a Status_Summary parity companion ALWAYS in sync, exported to HTML + PDF. Two-page format: page 1 = non-developer audience (team-specific), page 2 = software engineer summary. Auto-generated after every main Status update."

For every docs/<domain>/<integration>/Status.md (§11.4.45) a companion Status_Summary.md MUST exist with: (1) §11.4.44 revision header, (2) **Page 1 — For the** — audience- specific heading (audio team for docs/dolby/*, video team for docs/video/*, etc.) — plain-language summary, What works (1-3 bullets), What's broken or pending (1-3 bullets), Operator / team actions if any. NO code references, NO §-letter jargon, NO captured-evidence file paths, NO gate / mutation names. (3) **Page 2 — For software engineers** — §-letter references, gate names, commit hashes, captured-evidence paths, ATM-NNN cross-references. HTML + PDF exports travel with the markdown.

Generator: scripts/testing/generate_status_summary.sh <Status.md path> produces both pages. Invoked from scripts/testing/sync_integration_status.sh (§11.4.45 sync wrapper) on every Status.md update.

Composes with §11.4.45 (Status.md per integration — Status_Summary.md COMPLEMENTS, never replaces), §11.4.12 + §11.4.53 (parity discipline), §11.4.44 (revision header), §11.4.23 (colorizer for tracked-item references on page 2), §12.10 (CONTINUATION.md — non-developer stakeholders read Status_Summary; engineers read Status + CONTINUATION + Issues).

Pre-build gates CM-STATUS-SUMMARY-EXISTS-FOR-EVERY-STATUS + CM-STATUS-SUMMARY-TWO-AUDIENCE + CM-STATUS-SUMMARY-REVISION-HEADER + CM-COVENANT-114-56-PROPAGATION. Paired mutations delete a Status_Summary.md, remove the Page 1 heading, strip the anchor literal. No escape hatch — no --skip-status-summary, --engineer-only, --no-audience-split flag.

Canonical authority: constitution submodule Constitution.md §11.4.56.

Non-compliance is a release blocker regardless of context.

§11.4.57 — README.md doc-link section + revision metadata (User mandate, 2026-05-19)

Forensic anchor — verbatim user mandate (2026-05-19):

"Add a doc-link section to README.md — links to Issues + Issues_Summary + Fixed + Fixed_Summary + CONTINUATION + ALL Status docs + their exports. Each link shows revision + last-modified."

Every project's top-level README.md MUST contain a section titled Tracked-Items + Status Documents (heading MUST contain literal Tracked-Items). The section is a markdown table with columns: Document, Last modified (ISO 8601 UTC from §11.4.44 header), Revision (integer from same header), Markdown link, HTML link, PDF link. The section MUST link to: Issues.md + Issues_Summary.md (§11.4.12, §11.4.15, §11.4.16), Fixed.md + Fixed_Summary.md (§11.4.19, §11.4.53), CONTINUATION.md (§12.10), every docs/**/Status.md + its Status_Summary.md pair (§11.4.45, §11.4.56). Status docs are auto-discovered by the generator via `find docs -name 'Status.md'`.

Generator: `scripts/testing/update_readme_doc_links.sh` extracts each doc's Revision + Last modified, renders the markdown table, replaces the section between markers (`<!-- doc-link-section:begin -->/<!-- doc-link-section:end -->`) in README.md. Invoked from `sync_issues_docs.sh` (Issues / Fixed edits) AND `sync_integration_status.sh` (Status edits).

Composes with §11.4.12 + §11.4.19 + §11.4.53 (Issues / Fixed / summaries — README links all four), §11.4.44 (revision header data source), §11.4.45 + §11.4.56 (Status pairs enumeration), §12.10 (CONTINUATION.md explicit link).

Pre-build gates CM-README-DOC-LINK-SECTION-PRESENT (literal Tracked-Items + section markers) + CM-README-DOC-LINK-ROWS-COMPLETE (every canonical doc appears as a row) + CM-README-DOC-LINK-FRESHNESS (Last modified matches source within sync-wrapper granularity) + CM-COVENANT-114-57-PROPAGATION. Paired mutations strip the markers, remove the CONTINUATION row, backdate a Last modified cell, strip the anchor literal. No escape hatch — no --skip-readme-doc-links, --collapse-status-rows, --no-freshness-check flag.

Canonical authority: constitution submodule Constitution.md §11.4.57.

Non-compliance is a release blocker regardless of context.

§11.4.58 — Parallel-development methodology (User mandate, 2026-05-19)

Forensic anchor — direct user mandate (verbatim, 2026-05-19T~05:00Z MSK):

"We MUST DO one comprehensive research and planning in the background in parallel with current mainstream work: our current methodology of the development is very slow. ... We MUST CREATE adjusted improved version of working methodology where multiple workable items could be done in parallel, all come into the central (main) branch as they are done, then we at particular moment rebuild and reflash the System and in background full testing with validation and verification is done. Each parallel work on one workable (or more) item(s) must use parallel agents as much as possible! ... Writing in depth tests (all supported types of the tests) with Challenges and full HelixQA use is MANDATORY! Every test we execute besides executed with success MUST RESULT in proof that actual functionality being tested REALLY DOES WORK with NO BLUFF of any kind! Heavt enforcement of no-bluff / anti-bluff policy IS MANDATORY!"

Project work proceeds through the **Parallel Work Unit (PWU) pipeline** rather than sequential phase-chain. Each PWU is a self-contained workable item with mandatory components: ATM-NNN identifier per §11.4.54, Issues.md entry per §11.4.15+§11.4.16, file-scope manifest, §11.4.43 RED test, source patch, pre-build gate per §11.4.4(b) layer 1, post-flash test per §11.4.4(b) layer 3, paired §1.1 meta-test mutation, §11.4.4(b) layer 4 HelixQA Challenge bank entry, captured-evidence directory per §11.4.5+§11.4.52.

5-stage pipeline: (Stage 1 DEVELOP — parallel PWU agents in isolated worktrees) → (Stage 2 MERGE — serial conductor via `commit_all.sh` flock

- §11.4.41 4-step merge-first) → (Stage 3 REBUILD+FLASH — parallel where hardware allows) → (Stage 4 VALIDATE — parallel on D3+D4+meta-test+coverage) → (Stage 5 SWEEP — parallel HelixQA Challenges + Fixed.md migration + README refresh). Stage 1 of round N+1 overlaps with Stages 4-5 of round N — the throughput multiplier.

Synchronization mechanism: 4-layer lock hierarchy (L1 parent flock / L2 per-submodule git ops / L3 contention-path advisory locks for the 10 forbidden cross-PWU paths / L4 per-PWU git worktree). Disjoint-scope PWUs run fully parallel. Conflict detection at Stage 2: `git fetch --all --prune --tags + scope-overlap check` → overlap detected rejects PWU back to Stage 1.

Anti-bluff enforcement at merge time (all four required): C1 §11.4.43 RED-test captured-evidence (proves test catches regression), C2 §1.1 paired meta-test mutation (proves gate is not bluff), C3 §11.4.50 deterministic-consistency (3 or 10 iterations identical), C4 §11.4.5 captured-evidence (audio WAV / video screen recording / UI uiautomator dump / HelixQA `result.json`). Metadata-only / configuration-only / absence-of-error / `grep-without-runtime` PASS all REJECTED.

HelixQA mandatory. Every user-visible PWU MUST add a Challenge entry in `tools/helixqa/banks/atmosphere.yaml` referencing the PWU's ATM-NNN + dispatching to its on-device test + scoring PASS only on positive captured evidence per §11.4.5.

Pre-build gates CM-PWU-LOCK-HIERARCHY + CM-PWU-ANTI-BLUFF-COVERAGE + CM-PWU-MERGE-QUEUE-DISCIPLINE + CM-PWU-PARALLEL-AGENT-LIMIT + CM-COVENANT-114-58-PROPAGATION (anchor literal across 42 consumer files). Paired mutations strip the lock helpers / forge a READY marker without evidence / bypass the merge queue / spawn a 7th concurrent agent / strip the anchor literal — every mutation FAILs its gate.

No escape hatch — no `--skip-merge-queue`, `--allow-bypass`, `--no-anti-bluff-check`, `--unlimited-agents`, `--sequential-phase-chain-` mode flag. Composes with §11.4.4, §11.4.5, §11.4.6, §11.4.9, §11.4.15, §11.4.16, §11.4.19, §11.4.33, §11.4.41, §11.4.42, §11.4.43, §11.4.45, §11.4.49, §11.4.50, §11.4.52, §11.4.54, §11.4.57, §12.6, §12.7, §12.8, §12.10, §9.2.

Canonical authority: constitution submodule Constitution.md §11.4.58. Project-specific implementation reference in `consumer-side docs/guides/ PARALLEL_DEVELOPMENT_METHODODOLOGY.md`.

Non-compliance is a release blocker regardless of context.

§11.4.59 — README always-sync mandate (User mandate, 2026-05-19)

`README.md` is a §11.4.12-class always-sync document: HTML + PDF exports refresh on every update via `scripts/testing/sync_readme_export.sh` (`pandoc + weasyprint`); auto-invoked by `sync_issues_docs.sh` so a single doc-sync run refreshes `Issues / Issues_Summary / Fixed / Fixed_Summary / CONTINUATION / README` (`md + html + pdf`). `README` carries a §11.4.44 revision header and a Documentation Map section linking to every `Status.md + Status_Summary.md + spec + plan + guide + script-companion doc + changelog` + the constitution submodule, plus per-audience navigation. Pre-build gate `CM-README-EXPORT-SYNC` enforces mtime parity (`README.html + README.pdf ≥ README.md`). Paired meta-test mutation backdates HTML+PDF → gate FAILs. No escape hatch — no `--skip-readme-sync`, `--no-readme-export`, `--readme-stale-OK` flag. Composes with §11.4.12 + §11.4.18 + §11.4.44 + §11.4.45 + §11.4.53 + §11.4.56 + §11.4.57 + §12.10.

Canonical authority: constitution submodule Constitution.md §11.4.59.

Non-compliance is a release blocker regardless of context.

§11.4.60 — Documentation always-sync composite covenant (User mandate, 2026-05-19)

Eight documentation classes (`Issues`, `Issues_Summary`, `Fixed`, `Fixed_Summary`, `CONTINUATION`, `README`, every `Status.md`, every `Status_Summary.md`) MUST be in sync at all times across `.md + .html + .pdf` artefacts. Per-class anchors §11.4.12 / §11.4.44 / §11.4.45 / §11.4.53 / §11.4.56 / §11.4.57 / §11.4.59 / §12.10 govern individually; §11.4.60 binds them via single composite gate `CM-DOCS-COMPOSITE-SYNC` that FAILs if ANY instance's `.html` or `.pdf` mtime is older than `.md` mtime. Walks `docs/**` recursively for `Status.md` fleet. Paired mutation backdates `docs/Issues.html` → gate FAILs. No escape hatch — no `--skip-composite-doc-sync`, `--allow-stale-html`, `--summary-not-applicable` flag exists. Composes with §11.4.12 + §11.4.15 + §11.4.16 + §11.4.19 + §11.4.23 + §11.4.33 + §11.4.44 + §11.4.45 + §11.4.53 + §11.4.56 + §11.4.57 + §11.4.59 + §12.10.

Canonical authority: constitution submodule Constitution.md §11.4.60.

Non-compliance is a release blocker regardless of context.

§11.4.63 — Workable-items procedure docs as single source of truth (User mandate, 2026-05-19)

Every workable-item action (open / update / close / reopen / migrate) MUST follow the canonical procedure document at docs/procedures/issues/<Action>.md. Closed-set of 5 procedure docs: Creation.md, Updating.md, Resolution.md, Reopening.md, Migration.md. Each carries §11.4.44 revision header + HTML + PDF exports (refreshed by scripts/testing/sync_procedure_docs_export.sh, invoked as the final stage of sync_issues_docs.sh). ALL work — new features, behavioural changes, tasks (refactor / doc / infra / gate / audit / cleanup), bug fixes, investigation, documentation — MUST flow through these procedures. No ad-hoc procedure permitted.

Pre-build gate CM-PROCEDURES-DOCS-PRESENT checks all 5 procedure docs exist + carry revision header + have current HTML/PDF exports + sync helper is wired into sync_issues_docs.sh. Paired mutation: rename one procedure doc → gate FAILs. Propagation gate CM-COVENANT-114-63-PROPAGATION enforces this anchor literal in every CLAUDE.md / AGENTS.md.

Canonical authority: constitution submodule Constitution.md §11.4.63.

Non-compliance is a release blocker regardless of context.

MANDATORY HOST-SESSION SAFETY (Constitution §12)

Every script, test, helper, and AI agent MUST respect host-session safety. Non-compliance is a release blocker.

Forbidden — directly OR indirectly

1. Suspending the host (systemctl suspend, etc.).
2. Hibernating / hybrid-sleeping the host.
3. Logging out the user (loginctl terminate-session, pkill -u <user>, anything targeting user@<uid>.service).
4. Unbounded-memory operations inside user@<uid>.service cgroup — any command expected to exceed ~4 GiB RSS MUST be wrapped in a bounded execution scope.
5. Programmatic rfkill toggles, lid-switch handlers, power-button handlers — these cascade into idle-actions.
6. Disabling session managers "to make things faster".

Required safeguards for heavy scripts

1. Source the project's host-safety helper library.
2. Call its pre-flight check and abort if it fails.
3. Wrap any subprocess expected to exceed ~4 GiB RSS in a bounded execution scope.
4. Cap parallelism to fit available RAM.

§12.6 Memory-Budget Ceiling — 60% MAXIMUM

Forensic anchor — verbatim user mandate:

"First make sure that whatever we do through our procedures related to this project **MUST NOT** use more than 60% of total system memory! All processes **MUST** be able to function normally!"

Project procedures **MUST NOT** use more than **60%** of total system RAM. The remaining 40% is reserved for the operator's other workloads. No escape hatch — bypassing this is the bluff §11.4 forbids.

§12.10 Continuation document maintenance

docs/CONTINUATION.md **MUST** exist at the project root and reflect the live state of the work. Every non-trivial state change updates it in the same commit. Any agent must be able to resume work exactly where the previous session left off by reading this single file.

MANDATORY ABSOLUTE DATA SAFETY — ZERO RISK (Constitution §9)

Every destructive repository operation (history rewrite, force-push, branch deletion, bulk file removal, submodule de-init, object pruning) **MUST** follow the full §9 safety protocol **WITHOUT EXCEPTION**:

1. **Backup first, always** — hardlinked mirror of .git to a sibling backup directory (cp -al .git <backup>/repo.git.mirror) is near-instant and uses zero additional disk.
2. **Record metadata** — refs, tags, submodule pointers, HEAD commit, HEAD tree hash, tree content sha256.
3. **Define expected post-op state.**
4. **Run the destructive operation** — never with --no-verify, never with --force that bypasses hooks, never auto-force on failure.
5. **Post-op gate** — HEAD tree identical, all tags preserved, all submodule pointers intact, per-entry archive integrity 100%, pre-build gates green. If any check fails → restore immediately.
6. **Force-push authorization** — force-push is **NEVER** automatic. Each force-push event requires explicit human authorization **AND** requires the post-op gate to have passed.
7. **Audit trail** — every history rewrite gets a docs/changelogs/ "Force-push audit" section.

Hardlinked backup is so cheap (zero disk, <2 s) that there is **NEVER** an excuse to skip it.

MANDATORY COMMIT & PUSH CONSTRAINTS

1. **Use the project's official commit wrapper** for the main repo (e.g. scripts/commit_all.sh).
2. **NEVER use git add, git commit, or git push directly** on the main repo unless the project Constitution explicitly carves out a use case.

3. **Multi-upstream push is the norm** — every commit pushed to ALL configured upstream remotes. The constitution submodule ships an `install_upstreams.sh` (and an `Upstreams/` directory) that configures all remotes locally.
4. **NEVER skip hooks** (`--no-verify`, `--no-gpg-sign`) unless the user explicitly authorizes it for the session.

MANDATORY TESTING CONSTRAINTS

Tests **MUST** be run at every stage **WITHOUT EXCEPTION**:

1. **Pre-build / pre-merge verification** — BEFORE every build.
2. **Post-build / packaging verification** — AFTER every build.
3. **Post-deploy verification** — AFTER every deploy / flash.

NEVER skip tests. NEVER mark a test as "broken — disable for now" without fixing the underlying issue. NEVER ship a release with unresolved WARNs.

Code conventions

Universal conventions applicable to most projects:

- Use the project's preferred language for new code. Don't mix languages in one module without explicit Constitution permission.
- Static analyzers / linters / type-checkers **MUST** run clean (zero warnings) before commit.
- Style is set by the project's formatter; do not hand-edit style in PRs.
- Public APIs are documented at the source.

When in doubt

- Read `constitution/Constitution.md` for the canonical text.
- Cross-reference the project's `CLAUDE.md` / `AGENTS.md` for project- specific extensions.
- If still unclear, ask the operator. Do NOT guess. Do NOT bluff.

§11.4.65 — Universal Markdown export mandate (User mandate, 2026-05-19)

Every Markdown document inside the project that is NOT part of an application or service's source-code tree **MUST** have synchronized `.html` and `.pdf` siblings. Includes: `project-root/*.md`, `docs/**/*.md`, `scripts/**/*.md` (doc-format companion docs), owned-submodule top-level `README.md` / `CLAUDE.md` / `AGENTS.md` / `CHANGELOG.md` and their `docs/**/*.md`, `constitution/**/*.md`, owned HelixQA submodules' equivalents. Excludes: `external/**`, `prebuilts/**`, `packages/modules/**`, `kernel-5.10/**`, `out/**`, `build/**`, application/service source-code trees, and third-party submodules NOT in the owned set. Every edit triggers regeneration via `scripts/testing/`

`sync_all_markdown_exports.sh` (pandoc HTML + weasyprint PDF, timeout 60 per file, capped at 500 candidates). HTML + PDF mtime MUST be $\geq$ source .md mtime at all times.

Pre-build gates CM-UNIVERSAL-MARKDOWN-EXPORT-SYNC + CM-COVENANT-114-65-PROPAGATION. Paired meta-test mutations. Composes with §11.4.12 / §11.4.18 / §11.4.23 / §11.4.44 / §11.4.45 / §11.4.53 / §11.4.59 / §11.4.60 / §11.4.63 / §11.4.64. No escape hatch — no `--skip-md-exports`, `--no-pdf-only`, `--md-export-not-applicable` flag.

Canonical authority: constitution submodule Constitution.md §11.4.65.

Non-compliance is a release blocker regardless of context.

§11.4.66 — Blocker-resolution interactive-clarification mandate (User mandate, 2026-05-19)

When any task is blocked (operator decision, hardware access, external authorization, ambiguous scope), the agent MUST: (1) research what's doable from the agent side without operator input; (2) calculate minimum-viable operator input; (3) construct 2–4 mutually-exclusive options with one marked "Recommended" and each stating what the agent does after that answer; (4) present via the platform's interactive question mechanism (AskUserQuestion on Claude Code) — NEVER free-text "what would you like?" for closed- set decisions; (5) after the answer, resume work without follow-up round-trips. Composes with §11.4.6 / §11.4.7 / §11.4.40 / §11.4.41 / §11.4.42 / §11.4.52. No silent waiting; no bulk-text questions when interactive options would do.

Pre-build gate CM-COVENANT-114-66-PROPAGATION enforces the anchor literal across the 42-file consumer fleet. Paired meta- test mutation strips the literal → gate FAILs. No escape hatch — no `--skip-ask`, `--silent-wait`, `--free-form-only` flag.

Canonical authority: constitution submodule Constitution.md §11.4.66.

Non-compliance is a release blocker regardless of context.

§11.4.67 — Shell-script target-shell-parseability mandate (User mandate, 2026-05-19)

Forensic anchor — direct user mandate (verbatim, 2026-05-19): "any issue we spot must be fixed, bash scripts as well if they are broken!"

- "Make sure that this is mandatory rule!"

Every shell script that may be invoked under a target shell other than the one in its shebang MUST parse cleanly under that target shell. Forensic incident: `device/rockchip/rk3588/tests/test_all_fixes.sh:114` used `bash-only exec >>(tee -a "$f") 2>&1` on a `sh script.sh` callsite — Android `mksh` parses the whole script BEFORE executing, so the runtime `[ -n "${BASH_VERSION:-}" ]` guard could not save it. Fixed by wrapping in `eval 'exec >>(tee ...) 2>&1'` so the parser sees only a string.

Closed-set scope: every tracked .sh under `device/rockchip/rk3588/tests/`, `scripts/`, `scripts/testing/` (and equivalent paths in owned submodules). OUT of scope: `external/`, `prebuilts/`, `packages/modules/`, `kernel-5.10/`, `out/`,

build/, scripts/legacy/. Mandatory invariants: (1) every in-scope script parses under `sh -n`; (2) bash-only constructs (`>(...`), `<(...`), `[[ ]]`, `<<<`, arrays, `{var^^}`, etc.) MUST be wrapped in `eval` OR guarded by bash-only loading; (3) shebangs honest — `#!/bin/bash` only if bash actually expected; (4) fix at source per §11.4.1, never at callsites. Composes with §11.4.1 / §11.4.4 / §11.4.6 / §11.4.50 / §11.4.51.

Pre-build gate `CM-SCRIPT-TARGET-SHELL-PARSEABLE` runs `sh -n` on every in-scope script. Propagation gate `CM-COVENANT-114-67-PROPAGATION` enforces the anchor literal across the 44-file consumer fleet. Paired mutations: inject bash-only outside `eval` → parse gate FAILs; strip 11.4.67 literal → propagation gate FAILs. No escape hatch — no `--skip-parseability-check`, `--bash-only-script`, `--runtime-guard-suffices` flag.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.67.

Non-compliance is a release blocker regardless of context.

§11.4.68 — Positive sink-side / downstream evidence mandate (User mandate, 2026-05-20)

A test asserting audio or video routing PASS MUST capture and verify positive sink-side or downstream evidence — never config-only, never metadata-only, never PCM-open-state-only. Closed enumeration of acceptable evidence: (1) sink-side codec-state showing non-empty codec matching the expected regex during playback (Arvus / Sonos / Yamaha REST API); (2) PCM `hw_ptr` delta strictly positive across the playback window; (3) ALSA ELD demonstrating negotiated channel count + format; (4) `ffprobe-on-captured-mp4` with non-zero frames matching expected codec; (5) recording-analyzer matched event per §11.4.2 / §11.4.5 timeline; (6) `tinycap` RMS amplitude above floor.

Failure modes specifically forbidden: `arvus_probe_present` returning 1 silently dropping the test to SKIP; empty codec-state treated as evidence; `dumpsys media.audio_flinger` policy-side device field used as proof PCM actually opened; config-XML-only PASS without runtime evidence. All four are the §11.4 PASS-bluff pattern materialised on D3 2026-05-20 when the test suite reported audio- routing PASS while the user heard nothing and Arvus Codec-In-Use was empty.

Mandatory protections: (1) sink-side library helpers expose `*_require_reachable` returning exit code 2 (OPERATOR-BLOCKED) when sink unreachable; (2) test harness propagates 2 to summary as OPERATOR-BLOCKED, release tags blocked while OPERATOR-BLOCKED exists; (3) audio-routing tests rewritten to capture ≥1 positive sink/downstream evidence per the enumeration above; (4) anti-stickiness post-stop — test re-probes sink and asserts codec-state transitioned to N.E. / 0-frames-delta. No escape hatch — no `--skip-sink-evidence`, `--allow-empty-codec`, `--sink-unreachable-is-pass`, `--metadata-only-suffices` flag.

Composes with §11.4.2 / §11.4.5 / §11.4.13 / §11.4.14 / §11.4.46 / §11.4.49 / §11.4.50 / §11.4.52. Pre-build gates `CM-COVENANT-114-68-PROPAGATION` + `CM-POSITIVE-SINK-EVIDENCE-PER-AUDIO-TEST` with paired meta-test mutations per §1.1.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.68.

Non-compliance is a release blocker regardless of context.

§11.4.69 — Universal sink-side positive-evidence taxonomy + mechanical enforcement (User mandate, 2026-05-20)

Forensic anchor — direct user mandate (verbatim, 2026-05-20):

"THIS MUST HAPPEN NEVER AGAIN!!! We MUST HAVE this all working! Not just for audio but for every single piece of the System!!! Proper full automation when executed with success MUST MEAN that manual testing will be as much positive at least regarding the success results! ... Solution MUST BE universal, generic that solves working flows for all System components and for all future and all existing projects! ... Everything we do MUST BE validated and verified with rock-solid proofs and anti-bluff policy enforcement and fulfillment!"

Universal generalisation of §11.4.68 (audio-specific) across every user-visible feature class. Closes the PASS-bluff pattern where tests reported green while end users hit broken features (2026-05-19→20 D3 audio "82/84 PASS" + empty Arvus Codec-In-Use).

The mandate. Every user-visible feature MUST map to one entry in the closed-set §11.4.69 sink-side evidence taxonomy (audio_output, audio_input, video_display, network_throughput, network_connectivity, bluetooth_a2dp, bluetooth_pair, touch_input, sensor, gpu_render, storage_read, storage_write, mediacodec_decode, mediacodec_encode, miracast, cast, boot_service, package_install, permission_grant, wifi_link, wifi_throughput, ethernet_link, display_topology, drm_playback, subtitle_render — open to additions). Every PASS for a feature in the taxonomy MUST cite a captured-evidence artefact path matching the required evidence shape.

Helper contracts (additive during grace; mandatory after 2026-06-19):

- `ab_pass_with_evidence` <description> <evidence_path> — the new canonical PASS helper. Verifies path exists AND non-empty; emits PASS: <description> [evidence: <path>].
- `ab_skip_with_reason` <description> <closed-set-reason> — reasons: `geo_restricted`, `operator_attended`, `hardware_not_present`, `topology_unsupported`, `network_unreachable_external`, `feature_disabled_by_config`. Forbids `network_unreachable_external` for any taxonomy feature with a sink-side probe.
- Bare `ab_pass` deprecated — WARN pre-grace, FAIL post-grace (2026-06-19).

Mechanical enforcement. Three pre-build gates + three paired §1.1 meta-test mutations:

- `CM-SINK-EVIDENCE-PER-FEATURE` — walks tests for # §11.4.69 FEATURE: <class> annotation + verifies taxonomy probe + `ab_pass_with_evidence` use.
- `CM-NO-FAIL-OPEN-SKIP` — audits sink-side probe helpers; FAILs if any code path converts empty/unreachable response to PASS-counting SKIP for a feature class with a sink-side probe.
- `CM-AB-PASS-WITH-EVIDENCE-EVERYWHERE` — pre-grace WARN, post-grace FAIL on bare `ab_pass` calls.

Composes with §11.4.1 (FAIL-bluffs forbidden), §11.4.2 (recorded-evidence), §11.4.5 (audio + video 5-layer quality), §11.4.6 (no-guessing), §11.4.13 (sink-side captured-evidence), §11.4.27 (no-fakes-beyond-unit), §11.4.50 (deterministic consistency), §11.4.52 (autonomous-validation), §11.4.68 (audio-specific sink-side — §11.4.69 is the universal generalisation).

No escape hatch — no `--skip-evidence`, `--config-only-pass`, `--allow-fail-open-skip`, `--legacy-ab-pass-permitted` flag. The discipline exists because the 2026-05-20 forensic incident demonstrated the failure: tests reported audio-routing PASS while the user heard nothing and the Arvus Codec-In-Use field was empty.

Propagation gate CM-COVENANT-114-69-PROPAGATION enforces this anchor literal across the ~44-file consumer fleet. Paired mutation strips the literal → gate FAILs.

Canonical authority: constitution submodule Constitution.md §11.4.69.

Non-compliance is a release blocker regardless of context.

§11.4.70 — Subagent-driven execution is the default (User mandate, 2026-05-20)

Forensic anchor — direct user mandate (verbatim, 2026-05-20):

"Always do if possible Subagent-driven! Add this into our root (constitution Submodule) Constitution.md, CLAUDE.md and AGENTS.md. This should be the default choice ALWAYS!"

When executing implementation plans authored via `superpowers:writing-plans` (or any equivalent task-decomposed execution flow), the **default execution model is subagent-driven** per `superpowers:subagent-driven-development`. Inline execution via `superpowers:executing-plans` is permitted ONLY when (a) the task is trivial AND fits in a single sub-300-line edit, OR (b) the operator explicitly requests inline execution at brainstorm-handoff time.

Subagents bring an isolated context window per task (no conductor context bloat), a structurally separated review seam (conductor reviews subagent output, eliminating self-review blind spots), parallel-PWU compatibility (§11.4.58 — subagents ARE the parallel work units), and resumability across operator absence (subagents resume from on-disk plan + spec inputs).

Composes with §11.4.4 (four-layer coverage), §11.4.6 (no-guessing — subagent's captured output IS the evidence), §11.4.42 (iteration discipline), §11.4.43 (TDD-fix), §11.4.50 (deterministic consistency), §11.4.51 (LIVE_ADB_FIRST), §11.4.58 (parallel- development PWU).

No escape hatch — `--inline-execution-required`, `--no-subagents`, `--monolithic-execution` are NOT permitted flags. Skipping subagent-driven for non-trivial work without recorded operator authorisation is itself a §11.4 PASS-bluff. Pre-build gate CM-COVENANT-114-70-SUBAGENT-DEFAULT-PROPAGATION enforces this anchor literal across the ~44-file consumer fleet. Paired meta-test mutation strips the literal → gate FAILs.

Canonical authority: constitution submodule Constitution.md §11.4.70.

Non-compliance is a release blocker regardless of context.

§11.4.71 — Pre-Push Fetch + Investigate + Integrate Mandate (User mandate, 2026-05-20)

Everyday-push variant of §11.4.41 (force-push merge-first). Before pushing to any upstream for any repository (main repo or submodule), the agent MUST follow the 5-step pre-push cycle: (1) `git fetch --all --prune --tags`, (2) `git pull --no-rebase <remote> <branch>` for each remote whose tip differs from local, (3) investigate the diff vs OUR previous HEAD by reading every foreign commit's body (what changed, why, how it affects OUR project), (4) integrate mandatory changes with full §11.4.4(b) four-layer anti-bluff test coverage producing REAL captured-evidence proofs, (5) THEN push to every configured remote in cascade order.

Composes with §11.4.41 (force-push case) + §11.4.26 / §11.4.32 / §11.4.37 / §11.4.40 / §11.4.42 / §11.4.43 / §11.4.4(b) / §11.4.5 / §11.4.6.

Applies to parent repo + constitution submodule + every owned submodule + every nested submodule + every HelixQA dependency. Audit-trail per push reconstructable from docs/changelogs/<tag>.md + per-repo `git log` evidence.

No escape hatch — no `--skip-fetch`, `--no-investigate`, `--fast-push`, `--trust-upstream` flag. Pre-build gate CM-COVENANT-114-71-PROPAGATION + paired mutation.

Canonical authority: constitution submodule Constitution.md §11.4.71.

Non-compliance is a release blocker regardless of context.

§11.4.72 — Audio Top-Priority Mandate (User mandate, 2026-05-20)

Direct user mandate (verbatim): "Make sure all fixes for audio are always top priority in main working stream!" The conductor (main working stream — Claude Code session, AI agent, or human operator) MUST treat audio fixes as the highest-priority class on the serial dispatch queue. Audio scope includes §EU HDMI rejection, §EM multichannel HDMI, §ET Arvus integration, §EV/§EW/§EX D3 audio defects, HiFi (Fix #74, #76), AC3 (Fix #106), ES8388 (Fix #103/§104/#105), multichannel LPCM (Fix #112), and every future audio-stack improvement.

Composes with §11.4.42 (iteration-discipline — audio sits at apex of priority order) + §11.4.58 (parallel-development PWU — background research subagents run concurrently but do NOT preempt audio on the main-stream serial dispatch queue).

Operative rule: any time the conductor faces a choice between dispatching an audio task vs a non-audio task on the SAME serial resource, the audio task wins. No escape hatch — no "but this non-audio task is faster" or "but this research is more interesting" override. Audio-stack regressions are user-perceptible and high-impact (D3 silent post-flash is a release blocker), while research and refactors can wait.

Canonical authority: constitution submodule Constitution.md §11.4.72.

Non-compliance is a process violation regardless of context.

§11.4.73 — Main-specification document versioning + revision discipline (User mandate, 2026-05-20)

Direct user mandate (verbatim): "Make sure everything we add now in previous and upcoming requests IS ALWAYS applied to the main specification — if we have one. Since all these are not major changes we could increase Specification version per change for secondary version instead of the primary. Primary version MUST BE increased for much bigger levels of changes! Document MUST BE updated ALWAYS to follow the versioning rules we are applying here + revision and other properties we have!"

Applies only when a project recognises a main specification document (e.g. docs/specs/**/specification.V<n>.md). Two-axis versioning: **primary** (V1/V2/V3/...) bumps for major rewrites (old primary archived to <spec-dir>/archive/); **secondary** (Revision) bumps for additive operator requirements, refinements, polish (matches the §11.4.61 metadata-table Revision integer). Every operator-mandated requirement MUST land in the spec as part of the work that implements it. Cross-doc propagation copies MUST reference the active spec file, not a stale archived version. Composes with §11.4.44, §11.4.61, §11.4.59, §11.4.65.

Canonical authority: constitution submodule Constitution.md §11.4.73.

Non-compliance is a release blocker.

§11.4.74 — Submodule-catalogue-first discovery + extend-don't-reimplement (User mandate, 2026-05-20)

Direct user mandate (verbatim): "We MUST ALWAYS check which already developed features / functionalities do exist as a part of our comprehensive Submodules catalogue located in vasic-digital and HelixDevelopment organizations on GitHub and GitLab both! Project MUST BE aware of all its existence so we do not implement same things multiple times. For any missing features we MUST IMPLEMENT them properly and extend those Submodules further!"

Before scaffolding any new module / package / helper / utility, the contributor (human or AI agent) MUST: (1) survey the vasic-digital + HelixDevelopment orgs on GitHub + GitLab; (2) reuse an existing Submodule when it covers $\geq 80\%$ of the functionality; (3) extend in-place via upstream PR when $80\%+$ matches but features are missing — never duplicate the 80% in-project; (4) document the survey result in the relevant tracker row with a Catalogue-Check: reuse|extend|no-match <org/repo>@<sha> line.

Every Submodule in the catalogue is subject to the same development-cycle rules (§11.4 anti-bluff, §1.1 paired mutations, §11.4.10 credentials, §11.4.61 metadata + ToC, §11.4.65 universal export, §11.4.73 spec versioning, §2.1 multi-mirror push, §3 propagation order).

Canonical authority: constitution submodule Constitution.md §11.4.74.

Non-compliance is a process violation; severe cases (duplicate implementation landed without catalogue check) are release blockers.